Improving the availability of waterbird data for environmental emergency planning and response

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Background

Introduction

The mandate of the Canadian Wildlife Service (CWS) includes the management of all Migratory Birds and Species at Risk under its jurisdiction during environmental emergencies (an uncontrolled or unexpected incident involving the release of a polluting substance into the environment that may result in an immediate or long-term harmful effect on the environment) and Pollution Incidents (an incident involving the release or deposit of a substance that is harmful to wildlife). If such an incident or emergency results or may result in an immediate and/or long-term harmful effect on the life or health of wildlife and/or their habitat, it may be deemed a Wildlife Emergency. Determining whether any specific incident should be deemed a Wildlife Emergency depends on factors including the scope and severity of the incident (e.g., numbers of animals or area of habitat impacted), the potential for impacts to Species at Risk.

Wildlife Response Plans to guide response in the case of an environmental emergency are required for construction, operation or decommissioning activities that may impact the environment, and may also be required for some projects as part of an environmental assessment process. It is recommended that strategic response plans that describe the likely activities required to respond to an emergency be developed in advance for specific higher-risk activities and/or areas. However, in order to develop such plans, managers need to be able to identify areas of higher risk or sensitivity to wildlife or their habitats, as well as the species at risk of being impacted.

Similarly, when responding to an environmental emergency, CWS requires information such as which habitats are particularly sensitive or of concern, the numbers of Migratory Birds and Species at Risk likely present in that area at that time of year to assess potential impacts on wildlife populations and effectively guide mitigation measures to the highest priority locations. An initial wildlife impact assessment needs to be conducted as early in the incident response as possible (within the first 24 hours, according to guidelines; ECCC 2022). Thus, it is critical for CWS and other wildlife managers to have good baseline information on the distribution and abundance of Wildlife in known or potential high-risk areas and sensitive habitats, and to have this information needs to be readily available and compiled into a single dataset.

Although pollution that is discharged into a terrestrial landscape can potentially impact birds, mammals, reptiles, and amphibians, the spread is more limited than in open water (e.g., large bodies of freshwater) and aquatic (e.g., creeks, rivers, marshes, fens and bogs) environments. As a result, waterfowl, shorebirds, grebes, terns/gulls and other species with similar life history requirements (hereafter waterbirds) tend to be at higher risk for exposure to pollutants in environmental emergencies. Having a strong baseline understanding of which species may use different aquatic habitat types will therefore aid in assessing risks if an environmental emergency occurs. Species that tend to congregate in large numbers, for example when breeding (e.g., colonial breeding terns) or staging and moulting (e.g., waterfowl, shorebirds), are of particularly high concern when planning for wildlife emergencies, as the discharge of even a low volume of a contaminant can impact very large numbers of individuals. Therefore, the identification of habitats and specific locations where

these congregations occur is also of high priority when planning strategic response plans for environmental emergencies.

Within the Prairie Region of CWS, environmental emergency preparedness is advanced within the prairie ecosystem regions, which have historically been the areas most likely to be impacted by wildlife emergencies. However, the boreal region is increasingly at risk of environmental emergencies from industrial developments (e.g., mines) and the associated transportation pressures (e.g., transport of goods by road or rail). While collection and interpretation of baseline data on landbirds is reasonably advanced within this region, such information is largely lacking for waterbirds.

Goal & Objectives

To support planning and decision-making, the report comprises several analytical objectives that will increase our baseline understanding of waterbird distribution within the region:

1. Compile all available data from surveys in or near wetland habitats in AB, SK and MB;
2. Determine appropriate methods to harmonize the data from different survey methods into a common data format;
3. Import these data into WildTrax, an open data platform widely used by ECCC and other organizations;
4. Determine how to remove upland surveys;
5. Produce a single dataset, comprising only surveys of wetland habitats, available for analysis;
6. Produce a short report summarizing the wetland habitat associations for each waterbird species;
7. Determine the extent of unknown species identifications in recordings, and recommend an approach to reduce the number of such identifications in the future

Deliverables

These objectives will result in the following deliverables:

1. A report describing methods used to harmonize data;
2. List of WildTrax projects added, including whether each has additional habitat data available;
3. A report describing methods used to remove surveys in upland habitat types from the dataset;
4. A harmonized dataset, comprising only surveys of wetland habitats, that can be used for analysis;
5. A report summarizing basic wetland habitat associations for each waterbird species (for species which are detected in sufficiently large numbers to support analysis);
6. A report determining the extent of unknown species identifications in recordings (e.g., unknown shorebird, unknown duck), and recommending an approach to reduce the number of such identifications in the future.

All code for this report and the supporting analyses is available in a GitHub repository at https://github.com/borealbirds/Waterbirds.

Data Compilation & Harmonization

The following section describes methods and results for data compilation and harmonization (Deliverables 1 and 2).

Datasets

A comprehensive survey of surveys for shorebirds, waterbirds, and marshbirds in or near wetland habitats in AB, SK, and MB was undertaken. We worked with external agencies to identify potential datasets that are not yet available in WildTrax (e.g., Ducks Unlimited Canada, Audubon Canada, Birds Canada, Canadian Wildlife Service). We also included surveys from other regions of Canada to inform subsequent analyses. We considered the following data types in our survey:

- Autonomous recording unit (ARU) surveys
- Point count survey
- Point count surveys with call playbacks
- Aerial surveys
- Plot search surveys
- Transect surveys

Thirteen datasets were compiled (Table 1).

Table 1: Datasets included in compilation and harmonization.

Type	Organization	Name	Description	Surveys	Observations
Aerial	CWS (NOR)	WLS1992	Scoter driven surveys in Labrador. Helicopter/fixed-wing and helicopter double observer	1139	1284
Aerial	CWS (NOR)	WLS2008-09	Scoter driven surveys in Labrador. Helicopter/fixed-wing and helicopter double observer	2073	2236
Aerial	CWS (NOR)	SES2017–19	Scoter driven surveys in northern NWT, Nunavut, northern MB and northern Quebec. Helicopter double observer	19006	20375
Aerial	US FWS	WBPHS	Aerial transect surveys for waterfowl	873814	1381031
Plot search	US FWS	FSMS	Duck populations, prairie potholes Area search	235527	645078
Point count	Birds Canada	BCMMP	BC Marsh Monitoring Programs	1509	18748
Point count	eBird	eBird	eBird stationary point counts less than 10 minutes	1000189	5682902
Point count	BU	DU_SFMN	Wetland bird point counts	366	8051
Plot search	CWS	PRISM - PRA	Quadrat shorebird surveys	3052	9952

Table 1: Datasets included in compilation and harmonization. *(continued)*

Type	Organization	Name	Description	Surveys	Observations
Point count	ECCC	YRBWS	Yukon Co-operative Roadside Breeding Waterfowl/Waterbird Survey (1991 – 2016). Stationary Count and Linear Transect Checklists	24907	98987
Aerial	ECCC	RLWS	Distribution and Abundance of Waterfowl in the Rasmussen Lowlands, Nunavut, 1994-1995. Aerial transect surveys for waterfowl	1214	2757
Modified transect	Birds Canada	BCCWS	BC coastal waterbirds survey	31198	409154
Modified transect	Birds Canada	ACCWS	Atlantic Canada coastal waterbirds survey	2964	22863

Harmonization to WildTrax

The selected datasets were formatted to the format of the WildTrax "main" report for harmonization with other existing datasets. Datasets of ARU or point count style surveys were uploaded to the ARU and point count sensors in WildTrax, respectively, except for the eBird dataset, which is too large for storage in WildTrax. Aerial and plot search surveys were harmonized to the same data schema; however, were stored as .Rdata objects instead of upload to the WildTrax platform to avoid data misinterpreation because there are no appropriate sensors available for those datasets. Those harmonized datasets are available for download from a Google shared drive at https://drive.google.com/drive/folders/1n7lU7vGcl94dbL6ctlcJ1VKwGl9i6b3L. Code produced for data harmonization is available in a public Github repository at https://github.com/borealbirds/WildTrax-dataHarmonization.

Harmonization

The new datasets were then harmonized with the "BAM Dataset" (Barker et al. 2015), a collection of over 1.5 million bird surveys from across Canada to produce a comprehensive dataset of bird surveys. Following the definition of waterbirds as waterfowl, shorebirds, and marsh birds (i.e., excluding colonial waterbirds), we filtered the dataset to 142 species in 10 bird familes (Table 2): Anatidae (ducks and geese), Scolopacidae (sandpipers), Charadriidae (plovers), Recurvirostridae (stilts and avocets), Haematopodidae (osytercatches), Phalarapodidae (phalaropes), Rallidae (rails and coots), Ardeidae (herons and bitterns), Podicipedidae (grebes), and Gaviidae (loons).

The resultant dataset comprised 4.82 million surveys: 1.62 M from the BAM dataset, 1.00 M from eBird, and 2.20 M new surveys added through the data compilation and harmonization process, mostly from aerial transect and plot search surveys. The total number of waterbird observations across those surveys was 2.90 M. There were less observations than surveys because the database retained upland surveys where waterbirds were likely undetected; we retained these surveys in the database because they remain important and useful for understanding waterbird habitat relationships and species distribution modelling.

Table 2: Percentage of observations in harmonized dataset by waterbird family and survey type.

Family	ARU	Aerial.transect	Plot.search	Point.count
Anatidae	1.06	39.73	16.23	19.82
Ardeidae	0.16	0.00	0.00	1.95
Charadriidae	0.09	0.00	0.00	4.14
Gaviidae	0.51	0.03	0.00	1.61
Haematopodidae	0.00	0.00	0.00	0.03
Podicipedidae	0.23	0.01	0.00	0.96
Rallidae	0.63	1.61	0.00	1.36
Recurvirostridae	0.01	0.00	0.00	0.12
Scolopacidae	2.47	0.02	0.00	7.21

Habitat data extraction

For habitat data extraction and subsequent analytical objectives, we filtered the harmonized dataset to just point-level surveys (i.e., ARU and point count surveys; 2.65 M surveys), for which habitat data can be readily extracted and compared across datasets.

We extracted several data sources of wetland habitat for each unique location in the point survey dataset using Google Earth Engine:

- JRC Global Surface Water Version 1.4 (Pekel et al. 2016; 30 m resolution) was extracted as the mean within a 2 km buffer and the mean within a 20 km buffer
  - occurrence: the frequency with which water was present
  - seasonality: the mean number of months water was present
  - recurrence: the frequency with which water returns from year to year between 1984 and 2022
- Global Lakes and Wetlands Datavase Version 2 (Lehner et al. 2024; 15 arc second resolution) was extracted as the point value
  - dryland: the percentage that is classified as non-wetland
  - wetlandclass: the dominant wetland class
- C-CAP Wetland Pontential (30 m resolution) was extracted as the mean within a 2 km buffer and the mean within a 20 km buffer
  - wetness: likelihood of wetness

Data Cleaning

The following section describes methods and results for data cleaning (Deliverables 3, 4 and 6).

Unknown Identification Treatment

Number of Unknowns

First, we compared the frequency of unknown detections between ARU and in-person point count surveys to understand the magnitude of detections that are unknown waterbirds in acoustic recordings. There were approximately 25 times as many unknowns in ARU data as in point count data; however, this may be because

unknowns are not consistently reported in point count protocols and/or go ignored by in-person observers due to the inability to pause the survey, as compared with an ARU observer. The majority of unknowns in both survey types were ducks and yellowlegs (Table 3). Unidentified scaup were also common in point count surveys, but not in ARU surveys because this pair of species is almost entirely nonvocal and therefore not well detected by ARUs. Unidentified shorebirds were also common in ARU surveys.

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## Percent of unknowns in point count data:  0.014 %
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## Percent of unknowns in ARU data:  0.359 %
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Table 3: Percentage of unknown observations in point count and ARU surveys by unknown label.

Unknown label	**ARU**	**PointCount**
Mallard / American Black Duck	1.77	0.00
Unidentified Common/Red-breasted Merganser	0.01	0.07
Unidentified Dabbling Duck	1.17	0.07
Unidentified Diving Duck	0.35	0.13
Unidentified Duck	47.04	64.38
Unidentified Goldeneye	0.28	3.03
Unidentified Goose	0.56	0.73
Unidentified Greater / Lesser Scaup	0.07	9.23
Unidentified Grebe	0.01	0.40
Unidentified Loon	1.36	0.79
Unidentified Shorebird	15.31	2.90
Unidentified Swan	0.04	0.73
Unidentified Teal	0.01	0.00
Unidentified Waterfowl	3.20	0.00
Unidentified Western/Clark's Grebe	0.01	0.00
Unidentified Yellowlegs	20.09	16.23
Unidentified quack	8.72	0.73
Unidentified Dowitcher	0.00	0.07
Unidentified Scoter	0.00	0.40
Unidentified Short-billed / Long-billed Dowitcher	0.00	0.13

Treatment of Unknowns

We then downloaded the "AI" report for HawkEars from WildTrax for as many recordings with unknown waterbird detections as available in the WildTrax system. HawkEars is a high-performance deep-learning classifier for Canada birds (Huus et al. 2025) and is run in WildTrax with a minimum score threshold of 0.3. For each unknown detection, we determined if there were any HawkEars detections in the same time window for a species in the potential group of candidate species. If so we replaced that unknown detection, with the HawkEars detection. If there were multiple species identified, we used the highest scoring one.

We downloaded the AI report for 89.6% of unknown identifications in ARU data. For those 89.6% unknown detections, we found matches with HawkEars for 26.9%. In other words, we reduced the proportion of unknowns in the overall dataset from 0.359% to 0.235%. The majority of replacements were of unidentified ducks and yellowlegs (Table 4); HawkEars was not very effective at replacing unidentified shorebird detections. The mean score of detections replaced by HawkEars was 0.69.

Table 4: Percentage of unknown observations in ARU surveys by unknown label before and after HawkEars replacement.

Unknown label	**Before**	**After**	**Mean score**
Mallard / American Black Duck	1.77	0.88	0.56
Unidentified Common/Red-breasted Merganser	0.01	0.01	NaN
Unidentified Dabbling Duck	1.17	0.11	0.96
Unidentified Diving Duck	0.35	0.03	0.49
Unidentified Duck	47.04	27.30	0.61
Unidentified Goldeneye	0.28	0.10	0.73
Unidentified Goose	0.56	0.33	0.71
Unidentified Greater / Lesser Scaup	0.07	0.07	0.60
Unidentified Loon	1.36	1.12	NaN
Unidentified Shorebird	15.31	14.15	0.50
Unidentified Swan	0.04	0.01	NaN
Unidentified Waterfowl	3.20	2.64	0.66
Unidentified Yellowlegs	20.09	10.93	0.63
Unidentified quack	8.72	7.87	0.65

Upland Survey Flyover Removal

Instead of removing all upland surveys from the dataset, which would remove data points useful for applications like understanding habitat relationships, we completed two separate tasks: 1. Included habitat information that can be used to filter the dataset to wetland observations only (see 'Habitat data extraction' above); 2. Removed waterbird observations from the dataset that were at upland locations and had a high probability of being a flyover observation. In other words, an observation of an individual or group of waterbirds that were flying over the survey area and were not actually using the habitat for nesting or foraging purposes.

We removed high probability flyover observations from the dataset by modeling the probability that an observation was a flyover relative to our extracted habitat covariates from a subset of the point count surveys in the dataset that included and differentiated flyover observations as part of the survey protocol. There were just over 40 thousand observations in the dataset used for modelling flyover probability across nine of the ten waterbird families included in the dataset (Table 5). The total frequency of flyover observations in that dataset was 4.09%.

Table 5: Percentage of observations in point count surveys that were flyovers by waterbird family.

Family	**Percent flyovers**
Anatidae	5.11
Ardeidae	12.38
Charadriidae	1.47
Gaviidae	6.98
Podicipedidae	10.46
Rallidae	6.32
Recurvirostridae	2.07
Scolopacidae	4.10

We explored several modelling approaches, including logistic regression with and without random intercepts for waterbird family (GLM, GLMM), generalized additive modeling with a random intercept for family

(GAM), and boosted regression trees with a fixed effect for family (BRT). We withheld ten percent of the dataset and compared the predictive performance of each modeling approach using the area under the curve of the receiver operating characteristic (ROC AUC) and the log loss likelihood (Likelihood) of the model, both for the training and the testing dataset (Table 6). Overall, the boosted regression tree showed some signs of overfitting via lower performance on the test dataset than the training dataset; however, the performance on the test dataset was still better than any of the other three model types.

Table 6: Predictive performance of four models for probability of waterbird flyover on training and testing datasets.

Model	**Dataset**	**AUC ROC**	**Likelihood**
BRT	test	0.879	0.120
BRT	train	0.958	0.090
GAM	test	0.835	0.142
GAM	train	0.828	0.140
GLM	test	0.746	0.158
GLM	train	0.744	0.153
GLMM	test	0.771	0.153
GLMM	train	0.772	0.149

The wetland covariates extracted with the larger, 20 km radius buffer were the explained the most variation in the boosted regression tree model, particularly the percent dryland variable (18.7% variation explained), with wetland class and waterbird family also explaining 10.7% and 7.4% respectively. The covariates extracted at the 2 km radius each explained about 4% of the total variation explained by the model.

We made predictions for the probability of flyover for each detection in the overall survey dataset using the boosted regression tree model. There were no observations of Haematopodidae (oystercatchers) in the flyover training dataset, so we set observations for that waterbird family to Scolopacidae (sandpipers). We used a threshold of 20% probability of detection to label detections as flyovers for several reasons:

1. There was a break in the distribution of probability estimates at 0.20;
2. A value of 0.20 maximized the precision and recall of the model;
3. Thresholding with 0.20 identified approximately 7% of the detections in the dataset as flyovers, which was similar to the proportion in the flyover training dataset.

We removed all waterbird observations over this threshold from the waterbird dataset.

Data Analysis

The following section describes methods and results for data analysis (Deliverable 5), including additional exploration of waterbird detection frequency.

Wetland Habitat Associations

We used the entire harmonized and cleaned point survey dataset to summarize habitat relationships for each species with the habitat data extracted with a 2km radius and wetland class. We were interested in a high-level summary of habitat relationships, not density per se, and so we converted the count information in the waterbird dataset to presence/absence and used logistic regression a basic habitat occurrence model. We modelled the probability that a species was reported on a given survey relative to the habitat covariates as well as second order polynomials for the hour of day and day of year. We set the intercept value for

wetland class to the upland category. For each model, we randomly selected one observation from each cell of a hexagonal grid with 1.86 km spacing to avoid pseudoreplication and overpresentation of highly sampled areas. The dataset for each species was therefore approximately 200,000 observations. We only modelled species with at least 200 detections.

Most wetland classes had a postive effect on waterbird occurrence, except peatland and streams (Figure 1, Appendix A). Wetland seasonality had a positive effect on waterbird occurrence, while seasonality had a negative effect and only for several species. Both the probability of dryland and the wetness withhin 2 km had mixed effects across species.

Figure 1. Significant coefficient estimates from waterbird occurrence models by waterbird Family..

Detection Occurrence Summary

In addition to the initial objectives and deliverables in the statement of work, we reported on one additional component of the waterbird database: detection occurrence. In particular, we were interested in informing which species are detected at a similar occurrence rate with ARU surveys as point count surveys. For each species in the dataset, we calculated the following metrics for ARU surveys and for point count surveys:

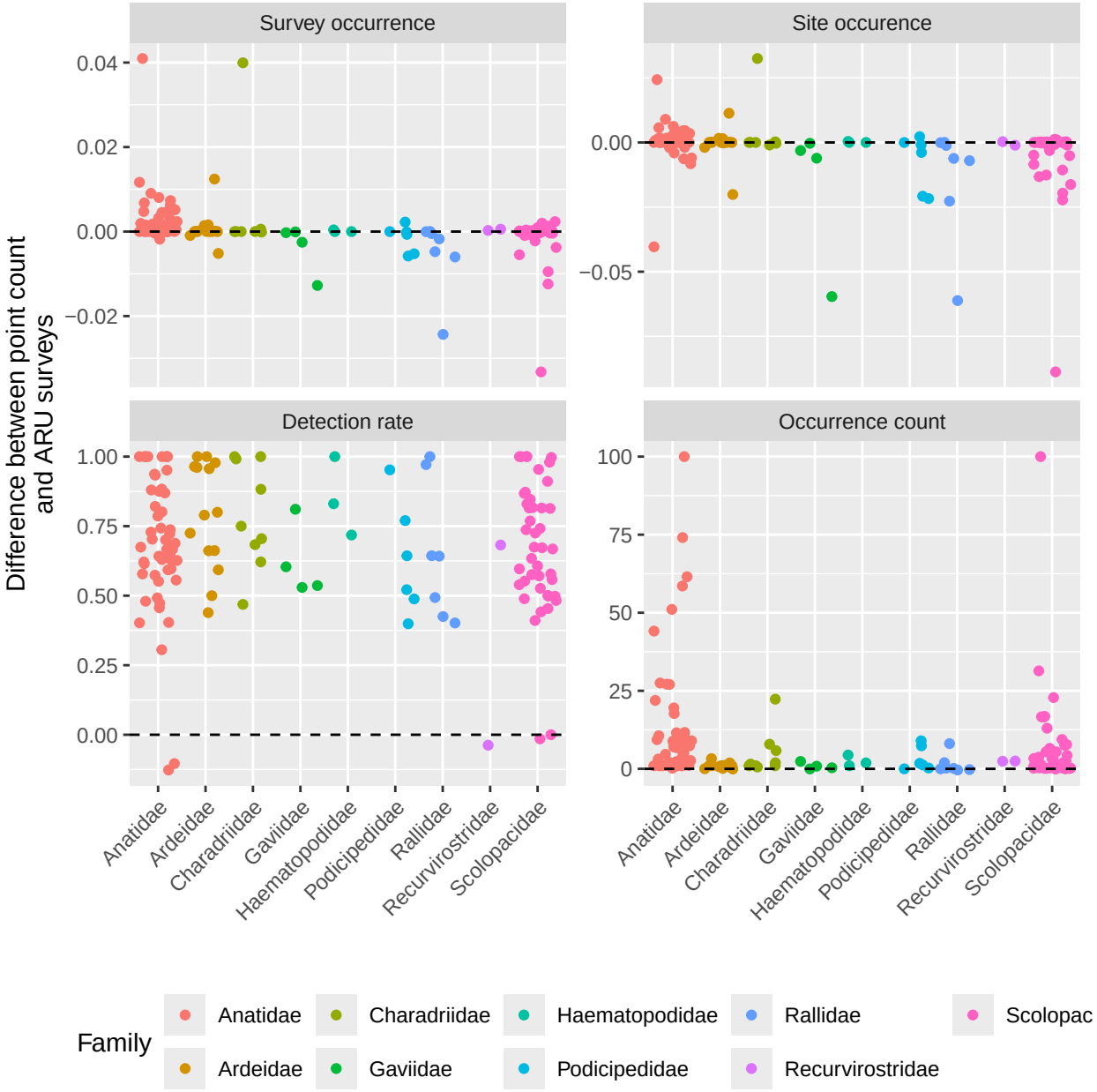
-Survey occurrence: the total proportion of surveys during which the species was detected; -Site occurrence: the proportion of sites at which the species was detected during at least one survey; -Detection rate: the proportion of surveys during which the species was detected for sites where it was detected during at least one survey; -Occurrence count: the mean count of the species when detected during a survey.

Our analysis did not account for differences in habitat quality or survey design between survey types and therefore assumes that the two methods have surveyed approximately the same populations across the large

dataset; however, the results should be interepreted as broad, preliminary results given this assumption. It should also be noted that some of the point count surveys, particularly the marsh monitoring program, use call playbacks for some species, that likely increased detection occurrence for the point count survey type.

Survey and site occurrence did not differ between method types on average; however, Anatids had higher survey occurrence on point count surveys than ARUs and Gavids, Rallids, and Podicipeds had higher survey and site occurence on ARUs than point count surveys (Figure 2, Appendix B). Detection rate was higher on point count surveys than ARUs; however, this is due to the single-visit nature of most point count survey datasets. Finally, the counts from point count surveys were higher than those from ARU surveys for almost all species, particularly for Anatids and Scolopacids. There were, however, several secretive marsh birds (e.g., yellow rails) that had higher counts on ARU surveys.

Figure 2. Difference between survey types (point count minus ARU) in ocurrence metrics of waterbirds by family.



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Appendix A - Habitat Occurrence Results

Table 7: Results from logistic regression of waterbird occurrence relative to wetland habitat. Results for intercept and temporal covariates are ommitted. The intercept value for wetland class was upland.

Family	Species	Covariate	Estimate	Standard error	P
Anatidae	American Black Duck	seasonality_2km	-1.068	0.245	0.000
Anatidae	American Black Duck	wetlandcoastal	2.900	0.153	0.000
Anatidae	American Black Duck	wetlandestuary	2.199	0.387	0.000
Anatidae	American Black Duck	wetlandlacustrine	1.147	0.188	0.000
Anatidae	American Black Duck	wetlandlake	1.455	0.151	0.000
Anatidae	American Black Duck	wetlandother	2.407	0.147	0.000
Anatidae	American Black Duck	wetlandpalustrine	1.312	0.175	0.000
Anatidae	American Black Duck	wetlandpeatland	0.295	0.139	0.034
Anatidae	American Black Duck	wetlandreservoir	1.846	0.308	0.000
Anatidae	American Black Duck	wetlandriver	1.324	0.135	0.000
Anatidae	American Black Duck	wetlandsaltmasch	3.649	0.167	0.000
Anatidae	American Black Duck	wetlandstream	0.647	0.320	0.043
Anatidae	American Black Duck	wetp_2km	-1.959	0.021	0.000
Anatidae	American Wigeon	seasonality_2km	1.464	0.196	0.000
Anatidae	American Wigeon	uplandp_2km	1.595	0.001	0.000
Anatidae	American Wigeon	wetlandcoastal	1.929	0.181	0.000
Anatidae	American Wigeon	wetlandephemeral	1.072	0.111	0.000
Anatidae	American Wigeon	wetlandestuary	1.941	0.590	0.001
Anatidae	American Wigeon	wetlandlacustrine	1.529	0.149	0.000
Anatidae	American Wigeon	wetlandlake	1.879	0.107	0.000
Anatidae	American Wigeon	wetlandother	1.958	0.125	0.000
Anatidae	American Wigeon	wetlandpalustrine	1.152	0.117	0.000
Anatidae	American Wigeon	wetlandpeatland	-0.589	0.102	0.000
Anatidae	American Wigeon	wetlandreservoir	2.238	0.275	0.000
Anatidae	American Wigeon	wetlandriver	1.352	0.075	0.000
Anatidae	American Wigeon	wetlandsaltmasch	2.286	0.242	0.000
Anatidae	American Wigeon	wetlandstream	-0.668	0.321	0.038
Anatidae	American Wigeon	wetp_2km	1.730	0.012	0.000
Anatidae	Barrow's Goldeneye	uplandp_2km	6.190	0.005	0.000
Anatidae	Barrow's Goldeneye	wetlandcoastal	1.317	0.451	0.004
Anatidae	Barrow's Goldeneye	wetlandephemeral	1.272	0.607	0.036
Anatidae	Barrow's Goldeneye	wetlandlacustrine	1.647	0.398	0.000
Anatidae	Barrow's Goldeneye	wetlandlake	2.329	0.227	0.000
Anatidae	Barrow's Goldeneye	wetlandother	3.231	0.213	0.000
Anatidae	Barrow's Goldeneye	wetlandpalustrine	2.172	0.238	0.000
Anatidae	Barrow's Goldeneye	wetlandreservoir	1.920	0.735	0.009
Anatidae	Barrow's Goldeneye	wetlandriver	1.934	0.217	0.000
Anatidae	Barrow's Goldeneye	wetlandsaltmasch	1.887	0.613	0.002
Anatidae	Barrow's Goldeneye	wetp_2km	-5.071	0.046	0.000
Anatidae	Blue-winged Teal	seasonality_2km	1.675	0.130	0.000
Anatidae	Blue-winged Teal	uplandp_2km	2.350	0.001	0.000

Table 7: Results from logistic regression of waterbird occurrence relative to wetland habitat. Results for intercept and temporal covariates are ommitted. The intercept value for wetland class was upland. *(continued)*

Family	**Species**	**Covariate**	**Estimate**	**Standard error**	**P**
Anatidae	Blue-winged Teal	wetlandcoastal	1.438	0.253	0.000
Anatidae	Blue-winged Teal	wetlandephemeral	1.610	0.069	0.000
Anatidae	Blue-winged Teal	wetlandlacustrine	2.206	0.113	0.000
Anatidae	Blue-winged Teal	wetlandlake	1.806	0.100	0.000
Anatidae	Blue-winged Teal	wetlandother	1.969	0.118	0.000
Anatidae	Blue-winged Teal	wetlandpalustrine	1.332	0.090	0.000
Anatidae	Blue-winged Teal	wetlandpeatland	-0.796	0.085	0.000
Anatidae	Blue-winged Teal	wetlandreservoir	1.908	0.343	0.000
Anatidae	Blue-winged Teal	wetlandriver	1.578	0.055	0.000
Anatidae	Blue-winged Teal	wetlandsaltmasch	1.203	0.455	0.008
Anatidae	Blue-winged Teal	wetlandstream	-0.810	0.271	0.003
Anatidae	Blue-winged Teal	wetp_2km	2.943	0.010	0.000
Anatidae	Bufflehead	occurrence_2km	-2.186	1.044	0.036
Anatidae	Bufflehead	seasonality_2km	2.073	0.459	0.000
Anatidae	Bufflehead	uplandp_2km	1.616	0.001	0.000
Anatidae	Bufflehead	wetlandcoastal	1.860	0.183	0.000
Anatidae	Bufflehead	wetlandephemeral	1.587	0.154	0.000
Anatidae	Bufflehead	wetlandlacustrine	2.372	0.132	0.000
Anatidae	Bufflehead	wetlandlake	2.532	0.107	0.000
Anatidae	Bufflehead	wetlandother	2.672	0.117	0.000
Anatidae	Bufflehead	wetlandpalustrine	1.949	0.118	0.000
Anatidae	Bufflehead	wetlandreservoir	2.400	0.264	0.000
Anatidae	Bufflehead	wetlandriver	1.746	0.096	0.000
Anatidae	Bufflehead	wetlandsaltmasch	1.843	0.287	0.000
Anatidae	Bufflehead	wetp_2km	-0.294	0.014	0.042
Anatidae	Cackling Goose	seasonality_2km	0.962	0.342	0.005
Anatidae	Cackling Goose	uplandp_2km	1.040	0.003	0.003
Anatidae	Cackling Goose	wetlandephemeral	1.104	0.231	0.000
Anatidae	Cackling Goose	wetlandlake	0.879	0.294	0.003
Anatidae	Cackling Goose	wetlandother	1.166	0.317	0.000
Anatidae	Cackling Goose	wetlandpalustrine	0.793	0.272	0.004
Anatidae	Cackling Goose	wetlandreservoir	1.536	0.732	0.036
Anatidae	Cackling Goose	wetlandriver	0.902	0.177	0.000
Anatidae	Cackling Goose	wetp_2km	1.808	0.029	0.000
Anatidae	Canada Goose	seasonality_2km	0.834	0.057	0.000
Anatidae	Canada Goose	uplandp_2km	-0.113	0.000	0.018
Anatidae	Canada Goose	wetlandcoastal	0.635	0.087	0.000
Anatidae	Canada Goose	wetlandephemeral	0.608	0.050	0.000
Anatidae	Canada Goose	wetlandestuary	0.930	0.230	0.000
Anatidae	Canada Goose	wetlandlacustrine	0.643	0.054	0.000
Anatidae	Canada Goose	wetlandlake	0.949	0.043	0.000
Anatidae	Canada Goose	wetlandother	0.908	0.059	0.000
Anatidae	Canada Goose	wetlandpalustrine	0.590	0.047	0.000
Anatidae	Canada Goose	wetlandpeatland	-0.329	0.030	0.000

Table 7: Results from logistic regression of waterbird occurrence relative to wetland habitat. Results for intercept and temporal covariates are ommitted. The intercept value for wetland class was upland. *(continued)*

Family	**Species**	**Covariate**	**Estimate**	**Standard error**	**P**
Anatidae	Canada Goose	wetlandreservoir	1.172	0.121	0.000
Anatidae	Canada Goose	wetlandriver	0.510	0.031	0.000
Anatidae	Canada Goose	wetlandsaltmasch	1.462	0.100	0.000
Anatidae	Canada Goose	wetlandstream	-0.264	0.090	0.003
Anatidae	Canada Goose	wetp_2km	1.379	0.005	0.000
Anatidae	Canvasback	seasonality_2km	1.344	0.263	0.000
Anatidae	Canvasback	uplandp_2km	2.938	0.003	0.000
Anatidae	Canvasback	wetlandcoastal	2.335	0.357	0.000
Anatidae	Canvasback	wetlandephemeral	1.506	0.152	0.000
Anatidae	Canvasback	wetlandlacustrine	2.856	0.205	0.000
Anatidae	Canvasback	wetlandlake	2.955	0.164	0.000
Anatidae	Canvasback	wetlandother	3.190	0.167	0.000
Anatidae	Canvasback	wetlandpalustrine	2.008	0.161	0.000
Anatidae	Canvasback	wetlandpeatland	-1.193	0.242	0.000
Anatidae	Canvasback	wetlandreservoir	3.693	0.363	0.000
Anatidae	Canvasback	wetlandriver	1.802	0.117	0.000
Anatidae	Canvasback	wetp_2km	2.711	0.019	0.000
Anatidae	Common Eider	uplandp_2km	-0.592	0.002	0.016
Anatidae	Common Eider	wetlandcoastal	5.138	1.011	0.000
Anatidae	Common Eider	wetlandestuary	4.566	1.103	0.000
Anatidae	Common Eider	wetlandother	5.668	1.006	0.000
Anatidae	Common Eider	wetlandpalustrine	3.041	1.057	0.004
Anatidae	Common Eider	wetlandpeatland	2.617	1.021	0.010
Anatidae	Common Eider	wetlandriver	2.256	1.049	0.032
Anatidae	Common Eider	wetlandsaltmasch	5.256	1.026	0.000
Anatidae	Common Eider	wetp_2km	-17.135	0.176	0.000
Anatidae	Common Goldeneye	wetlandcoastal	1.914	0.170	0.000
Anatidae	Common Goldeneye	wetlandephemeral	1.817	0.193	0.000
Anatidae	Common Goldeneye	wetlandestuary	2.922	0.291	0.000
Anatidae	Common Goldeneye	wetlandlacustrine	2.051	0.139	0.000
Anatidae	Common Goldeneye	wetlandlake	2.618	0.117	0.000
Anatidae	Common Goldeneye	wetlandother	2.411	0.131	0.000
Anatidae	Common Goldeneye	wetlandpalustrine	1.687	0.142	0.000
Anatidae	Common Goldeneye	wetlandreservoir	2.802	0.205	0.000
Anatidae	Common Goldeneye	wetlandriver	1.704	0.113	0.000
Anatidae	Common Goldeneye	wetlandsaltmasch	1.565	0.276	0.000
Anatidae	Common Goldeneye	wetp_2km	-1.504	0.017	0.000
Anatidae	Common Merganser	wetlandcoastal	1.862	0.184	0.000
Anatidae	Common Merganser	wetlandestuary	2.620	0.301	0.000
Anatidae	Common Merganser	wetlandlacustrine	1.957	0.153	0.000
Anatidae	Common Merganser	wetlandlake	2.666	0.131	0.000
Anatidae	Common Merganser	wetlandother	2.130	0.152	0.000
Anatidae	Common Merganser	wetlandpalustrine	1.567	0.164	0.000
Anatidae	Common Merganser	wetlandreservoir	2.822	0.214	0.000

Table 7: Results from logistic regression of waterbird occurrence relative to wetland habitat. Results for intercept and temporal covariates are ommitted. The intercept value for wetland class was upland. *(continued)*

Family	**Species**	**Covariate**	**Estimate**	**Standard error**	**P**
Anatidae	Common Merganser	wetlandriver	1.944	0.129	0.000
Anatidae	Common Merganser	wetlandsaltmasch	2.168	0.227	0.000
Anatidae	Common Merganser	wetlandstream	1.087	0.283	0.000
Anatidae	Common Merganser	wetp_2km	-2.679	0.020	0.000
Anatidae	Gadwall	seasonality_2km	1.570	0.136	0.000
Anatidae	Gadwall	uplandp_2km	2.702	0.002	0.000
Anatidae	Gadwall	wetlandcoastal	1.823	0.213	0.000
Anatidae	Gadwall	wetlandephemeral	1.562	0.071	0.000
Anatidae	Gadwall	wetlandestuary	1.961	0.717	0.006
Anatidae	Gadwall	wetlandlacustrine	1.739	0.141	0.000
Anatidae	Gadwall	wetlandlake	1.692	0.109	0.000
Anatidae	Gadwall	wetlandother	1.913	0.124	0.000
Anatidae	Gadwall	wetlandpalustrine	1.200	0.097	0.000
Anatidae	Gadwall	wetlandpeatland	-1.815	0.135	0.000
Anatidae	Gadwall	wetlandreservoir	2.565	0.264	0.000
Anatidae	Gadwall	wetlandriver	1.416	0.059	0.000
Anatidae	Gadwall	wetlandsaltmasch	2.459	0.273	0.000
Anatidae	Gadwall	wetlandstream	-1.492	0.381	0.000
Anatidae	Gadwall	wetp_2km	2.802	0.011	0.000
Anatidae	Greater Scaup	uplandp_2km	-0.927	0.003	0.001
Anatidae	Greater Scaup	wetlandcoastal	2.114	0.387	0.000
Anatidae	Greater Scaup	wetlandestuary	2.335	0.771	0.002
Anatidae	Greater Scaup	wetlandlacustrine	1.127	0.389	0.004
Anatidae	Greater Scaup	wetlandlake	2.029	0.318	0.000
Anatidae	Greater Scaup	wetlandother	2.264	0.339	0.000
Anatidae	Greater Scaup	wetlandpalustrine	1.030	0.418	0.014
Anatidae	Greater Scaup	wetlandreservoir	1.890	0.645	0.003
Anatidae	Greater Scaup	wetlandriver	0.764	0.322	0.018
Anatidae	Greater Scaup	wetlandsaltmasch	2.811	0.405	0.000
Anatidae	Greater Scaup	wetp_2km	-0.952	0.046	0.037
Anatidae	Greater White-fronted Goose	seasonality_2km	1.784	0.367	0.000
Anatidae	Greater White-fronted Goose	uplandp_2km	1.458	0.003	0.000
Anatidae	Greater White-fronted Goose	wetlandephemeral	0.412	0.208	0.048
Anatidae	Greater White-fronted Goose	wetlandlake	1.095	0.212	0.000
Anatidae	Greater White-fronted Goose	wetlandother	0.937	0.268	0.000
Anatidae	Greater White-fronted Goose	wetlandpeatland	-0.858	0.177	0.000
Anatidae	Greater White-fronted Goose	wetlandriver	0.819	0.130	0.000
Anatidae	Greater White-fronted Goose	wetlandstream	-1.405	0.712	0.048
Anatidae	Greater White-fronted Goose	wetp_2km	2.029	0.023	0.000
Anatidae	Green-winged Teal	seasonality_2km	1.128	0.170	0.000
Anatidae	Green-winged Teal	uplandp_2km	1.252	0.001	0.000
Anatidae	Green-winged Teal	wetlandcoastal	1.807	0.184	0.000
Anatidae	Green-winged Teal	wetlandephemeral	1.559	0.098	0.000
Anatidae	Green-winged Teal	wetlandlacustrine	1.402	0.149	0.000

Table 7: Results from logistic regression of waterbird occurrence relative to wetland habitat. Results for intercept and temporal covariates are ommitted. The intercept value for wetland class was upland. *(continued)*

Family	**Species**	**Covariate**	**Estimate**	**Standard error**	**P**
Anatidae	Green-winged Teal	wetlandlake	1.315	0.122	0.000
Anatidae	Green-winged Teal	wetlandother	1.732	0.135	0.000
Anatidae	Green-winged Teal	wetlandpalustrine	0.919	0.127	0.000
Anatidae	Green-winged Teal	wetlandpeatland	-0.311	0.093	0.001
Anatidae	Green-winged Teal	wetlandreservoir	1.339	0.389	0.001
Anatidae	Green-winged Teal	wetlandriver	1.311	0.076	0.000
Anatidae	Green-winged Teal	wetlandsaltmasch	2.613	0.203	0.000
Anatidae	Green-winged Teal	wetp_2km	1.616	0.012	0.000
Anatidae	Hooded Merganser	occurrence_2km	-2.308	1.038	0.026
Anatidae	Hooded Merganser	seasonality_2km	0.891	0.401	0.026
Anatidae	Hooded Merganser	uplandp_2km	0.699	0.002	0.000
Anatidae	Hooded Merganser	wetlandcoastal	1.859	0.219	0.000
Anatidae	Hooded Merganser	wetlandlacustrine	2.238	0.172	0.000
Anatidae	Hooded Merganser	wetlandlake	2.272	0.148	0.000
Anatidae	Hooded Merganser	wetlandother	1.624	0.195	0.000
Anatidae	Hooded Merganser	wetlandpalustrine	2.118	0.155	0.000
Anatidae	Hooded Merganser	wetlandpeatland	0.403	0.147	0.006
Anatidae	Hooded Merganser	wetlandreservoir	2.070	0.361	0.000
Anatidae	Hooded Merganser	wetlandriver	1.687	0.135	0.000
Anatidae	Hooded Merganser	wetlandsaltmasch	1.617	0.344	0.000
Anatidae	Hooded Merganser	wetlandstream	0.852	0.309	0.006
Anatidae	Hooded Merganser	wetp_2km	-0.630	0.020	0.001
Anatidae	Lesser Scaup	seasonality_2km	2.268	0.360	0.000
Anatidae	Lesser Scaup	uplandp_2km	2.492	0.002	0.000
Anatidae	Lesser Scaup	wetlandcoastal	1.939	0.254	0.000
Anatidae	Lesser Scaup	wetlandephemeral	1.413	0.129	0.000
Anatidae	Lesser Scaup	wetlandlacustrine	2.437	0.150	0.000
Anatidae	Lesser Scaup	wetlandlake	2.617	0.115	0.000
Anatidae	Lesser Scaup	wetlandother	2.729	0.128	0.000
Anatidae	Lesser Scaup	wetlandpalustrine	1.905	0.117	0.000
Anatidae	Lesser Scaup	wetlandpeatland	-0.627	0.134	0.000
Anatidae	Lesser Scaup	wetlandreservoir	2.780	0.307	0.000
Anatidae	Lesser Scaup	wetlandriver	1.709	0.089	0.000
Anatidae	Lesser Scaup	wetlandsaltmasch	1.715	0.460	0.000
Anatidae	Lesser Scaup	wetlandstream	-1.352	0.582	0.020
Anatidae	Lesser Scaup	wetp_2km	1.532	0.014	0.000
Anatidae	Long-tailed Duck	uplandp_2km	-2.104	0.002	0.000
Anatidae	Long-tailed Duck	wetlandcoastal	3.303	0.624	0.000
Anatidae	Long-tailed Duck	wetlandephemeral	2.369	1.159	0.041
Anatidae	Long-tailed Duck	wetlandlacustrine	1.602	0.647	0.013
Anatidae	Long-tailed Duck	wetlandlake	2.792	0.605	0.000
Anatidae	Long-tailed Duck	wetlandother	4.079	0.599	0.000
Anatidae	Long-tailed Duck	wetlandpalustrine	2.077	0.673	0.002
Anatidae	Long-tailed Duck	wetlandpeatland	2.553	0.592	0.000

Table 7: Results from logistic regression of waterbird occurrence relative to wetland habitat. Results for intercept and temporal covariates are ommitted. The intercept value for wetland class was upland. *(continued)*

Family	**Species**	**Covariate**	**Estimate**	**Standard error**	**P**
Anatidae	Long-tailed Duck	wetlandriver	1.601	0.623	0.010
Anatidae	Long-tailed Duck	wetlandsaltmasch	3.550	0.662	0.000
Anatidae	Long-tailed Duck	wetp_2km	-4.337	0.059	0.000
Anatidae	Mallard	seasonality_2km	0.800	0.054	0.000
Anatidae	Mallard	uplandp_2km	0.832	0.001	0.000
Anatidae	Mallard	wetlandcoastal	1.093	0.094	0.000
Anatidae	Mallard	wetlandephemeral	1.014	0.046	0.000
Anatidae	Mallard	wetlandestuary	1.523	0.245	0.000
Anatidae	Mallard	wetlandlacustrine	1.167	0.059	0.000
Anatidae	Mallard	wetlandlake	1.351	0.046	0.000
Anatidae	Mallard	wetlandother	1.348	0.061	0.000
Anatidae	Mallard	wetlandpalustrine	0.813	0.049	0.000
Anatidae	Mallard	wetlandpeatland	-0.668	0.036	0.000
Anatidae	Mallard	wetlandreservoir	1.634	0.129	0.000
Anatidae	Mallard	wetlandriver	0.928	0.031	0.000
Anatidae	Mallard	wetlandsaltmasch	1.566	0.120	0.000
Anatidae	Mallard	wetlandstream	-0.325	0.098	0.001
Anatidae	Mallard	wetp_2km	2.012	0.005	0.000
Anatidae	Northern Pintail	seasonality_2km	0.986	0.128	0.000
Anatidae	Northern Pintail	uplandp_2km	1.252	0.001	0.000
Anatidae	Northern Pintail	wetlandcoastal	2.135	0.181	0.000
Anatidae	Northern Pintail	wetlandephemeral	1.176	0.089	0.000
Anatidae	Northern Pintail	wetlandlacustrine	1.206	0.168	0.000
Anatidae	Northern Pintail	wetlandlake	1.435	0.128	0.000
Anatidae	Northern Pintail	wetlandother	1.748	0.141	0.000
Anatidae	Northern Pintail	wetlandpalustrine	0.536	0.139	0.000
Anatidae	Northern Pintail	wetlandpeatland	-1.230	0.120	0.000
Anatidae	Northern Pintail	wetlandreservoir	2.164	0.317	0.000
Anatidae	Northern Pintail	wetlandriver	1.062	0.072	0.000
Anatidae	Northern Pintail	wetlandsaltmasch	2.942	0.209	0.000
Anatidae	Northern Pintail	wetlandstream	-0.820	0.321	0.011
Anatidae	Northern Pintail	wetp_2km	2.952	0.012	0.000
Anatidae	Northern Shoveler	seasonality_2km	1.444	0.125	0.000
Anatidae	Northern Shoveler	uplandp_2km	2.887	0.002	0.000
Anatidae	Northern Shoveler	wetlandcoastal	1.756	0.221	0.000
Anatidae	Northern Shoveler	wetlandephemeral	1.556	0.071	0.000
Anatidae	Northern Shoveler	wetlandlacustrine	2.035	0.130	0.000
Anatidae	Northern Shoveler	wetlandlake	1.891	0.104	0.000
Anatidae	Northern Shoveler	wetlandother	2.066	0.118	0.000
Anatidae	Northern Shoveler	wetlandpalustrine	1.151	0.099	0.000
Anatidae	Northern Shoveler	wetlandpeatland	-1.148	0.104	0.000
Anatidae	Northern Shoveler	wetlandreservoir	2.746	0.258	0.000
Anatidae	Northern Shoveler	wetlandriver	1.563	0.057	0.000
Anatidae	Northern Shoveler	wetlandsaltmasch	2.560	0.275	0.000

Table 7: Results from logistic regression of waterbird occurrence relative to wetland habitat. Results for intercept and temporal covariates are ommitted. The intercept value for wetland class was upland. *(continued)*

Family	**Species**	**Covariate**	**Estimate**	**Standard error**	**P**
Anatidae	Northern Shoveler	wetlandstream	-1.267	0.337	0.000
Anatidae	Northern Shoveler	wetp_2km	2.880	0.010	0.000
Anatidae	Red-breasted Merganser	uplandp_2km	-0.801	0.002	0.000
Anatidae	Red-breasted Merganser	wetlandcoastal	2.403	0.340	0.000
Anatidae	Red-breasted Merganser	wetlandephemeral	2.486	0.653	0.000
Anatidae	Red-breasted Merganser	wetlandestuary	2.002	0.595	0.001
Anatidae	Red-breasted Merganser	wetlandlacustrine	1.121	0.369	0.002
Anatidae	Red-breasted Merganser	wetlandlake	2.155	0.317	0.000
Anatidae	Red-breasted Merganser	wetlandother	3.167	0.312	0.000
Anatidae	Red-breasted Merganser	wetlandpalustrine	1.567	0.374	0.000
Anatidae	Red-breasted Merganser	wetlandriver	1.247	0.334	0.000
Anatidae	Red-breasted Merganser	wetlandsaltmasch	2.998	0.366	0.000
Anatidae	Red-breasted Merganser	wetp_2km	-7.551	0.069	0.000
Anatidae	Redhead	seasonality_2km	1.646	0.268	0.000
Anatidae	Redhead	uplandp_2km	2.367	0.002	0.000
Anatidae	Redhead	wetlandcoastal	2.017	0.322	0.000
Anatidae	Redhead	wetlandephemeral	1.583	0.132	0.000
Anatidae	Redhead	wetlandlacustrine	2.148	0.209	0.000
Anatidae	Redhead	wetlandlake	2.637	0.145	0.000
Anatidae	Redhead	wetlandother	2.562	0.168	0.000
Anatidae	Redhead	wetlandpalustrine	1.802	0.144	0.000
Anatidae	Redhead	wetlandpeatland	-1.299	0.212	0.000
Anatidae	Redhead	wetlandreservoir	2.954	0.375	0.000
Anatidae	Redhead	wetlandriver	1.860	0.100	0.000
Anatidae	Redhead	wetlandsaltmasch	1.439	0.719	0.045
Anatidae	Redhead	wetp_2km	2.448	0.017	0.000
Anatidae	Ring-necked Duck	seasonality_2km	1.123	0.300	0.000
Anatidae	Ring-necked Duck	uplandp_2km	1.417	0.001	0.000
Anatidae	Ring-necked Duck	wetlandcoastal	1.156	0.278	0.000
Anatidae	Ring-necked Duck	wetlandephemeral	1.068	0.189	0.000
Anatidae	Ring-necked Duck	wetlandestuary	1.544	0.721	0.032
Anatidae	Ring-necked Duck	wetlandlacustrine	2.021	0.153	0.000
Anatidae	Ring-necked Duck	wetlandlake	2.374	0.117	0.000
Anatidae	Ring-necked Duck	wetlandother	1.867	0.156	0.000
Anatidae	Ring-necked Duck	wetlandpalustrine	2.152	0.119	0.000
Anatidae	Ring-necked Duck	wetlandpeatland	0.438	0.109	0.000
Anatidae	Ring-necked Duck	wetlandreservoir	2.217	0.310	0.000
Anatidae	Ring-necked Duck	wetlandriver	1.605	0.104	0.000
Anatidae	Ring-necked Duck	wetlandsaltmasch	1.812	0.321	0.000
Anatidae	Ruddy Duck	seasonality_2km	2.499	0.507	0.000
Anatidae	Ruddy Duck	uplandp_2km	4.794	0.003	0.000
Anatidae	Ruddy Duck	wetlandcoastal	1.745	0.517	0.001
Anatidae	Ruddy Duck	wetlandephemeral	1.522	0.171	0.000
Anatidae	Ruddy Duck	wetlandlacustrine	3.384	0.190	0.000

Table 7: Results from logistic regression of waterbird occurrence relative to wetland habitat. Results for intercept and temporal covariates are ommitted. The intercept value for wetland class was upland. *(continued)*

Family	**Species**	**Covariate**	**Estimate**	**Standard error**	**P**
Anatidae	Ruddy Duck	wetlandlake	3.130	0.159	0.000
Anatidae	Ruddy Duck	wetlandother	3.407	0.161	0.000
Anatidae	Ruddy Duck	wetlandpalustrine	2.178	0.159	0.000
Anatidae	Ruddy Duck	wetlandpeatland	-1.034	0.239	0.000
Anatidae	Ruddy Duck	wetlandreservoir	2.762	0.595	0.000
Anatidae	Ruddy Duck	wetlandriver	2.147	0.117	0.000
Anatidae	Ruddy Duck	wetp_2km	1.940	0.019	0.000
Anatidae	Snow Goose	seasonality_2km	0.969	0.207	0.000
Anatidae	Snow Goose	wetlandcoastal	1.258	0.208	0.000
Anatidae	Snow Goose	wetlandephemeral	0.855	0.143	0.000
Anatidae	Snow Goose	wetlandestuary	2.917	0.307	0.000
Anatidae	Snow Goose	wetlandother	0.797	0.191	0.000
Anatidae	Snow Goose	wetlandpalustrine	0.422	0.169	0.013
Anatidae	Snow Goose	wetlandpeatland	-0.638	0.121	0.000
Anatidae	Snow Goose	wetlandreservoir	1.302	0.373	0.000
Anatidae	Snow Goose	wetlandriver	0.461	0.107	0.000
Anatidae	Snow Goose	wetlandsaltmasch	1.798	0.240	0.000
Anatidae	Snow Goose	wetp_2km	1.787	0.016	0.000
Anatidae	Surf Scoter	wetlandcoastal	3.606	0.623	0.000
Anatidae	Surf Scoter	wetlandephemeral	2.864	1.160	0.014
Anatidae	Surf Scoter	wetlandestuary	2.385	1.166	0.041
Anatidae	Surf Scoter	wetlandlake	2.557	0.618	0.000
Anatidae	Surf Scoter	wetlandother	4.503	0.598	0.000
Anatidae	Surf Scoter	wetlandpalustrine	2.310	0.682	0.001
Anatidae	Surf Scoter	wetlandreservoir	2.560	0.922	0.005
Anatidae	Surf Scoter	wetlandriver	1.618	0.661	0.014
Anatidae	Surf Scoter	wetlandsaltmasch	3.372	0.702	0.000
Anatidae	Surf Scoter	wetp_2km	-8.311	0.114	0.000
Anatidae	Trumpeter Swan	seasonality_2km	0.590	0.278	0.034
Anatidae	Trumpeter Swan	uplandp_2km	0.672	0.002	0.000
Anatidae	Trumpeter Swan	wetlandlacustrine	1.977	0.183	0.000
Anatidae	Trumpeter Swan	wetlandlake	2.060	0.152	0.000
Anatidae	Trumpeter Swan	wetlandother	1.133	0.232	0.000
Anatidae	Trumpeter Swan	wetlandpalustrine	0.953	0.207	0.000
Anatidae	Trumpeter Swan	wetlandpeatland	0.521	0.129	0.000
Anatidae	Trumpeter Swan	wetlandreservoir	2.390	0.318	0.000
Anatidae	Trumpeter Swan	wetlandriver	1.369	0.129	0.000
Anatidae	Tundra Swan	occurrence_2km	-1.544	0.737	0.036
Anatidae	Tundra Swan	uplandp_2km	-0.533	0.002	0.012
Anatidae	Tundra Swan	wetlandcoastal	0.731	0.319	0.022
Anatidae	Tundra Swan	wetlandlacustrine	0.598	0.239	0.012
Anatidae	Tundra Swan	wetlandlake	1.090	0.188	0.000
Anatidae	Tundra Swan	wetlandother	0.921	0.240	0.000
Anatidae	Tundra Swan	wetlandpeatland	-0.523	0.147	0.000

Table 7: Results from logistic regression of waterbird occurrence relative to wetland habitat. Results for intercept and temporal covariates are ommitted. The intercept value for wetland class was upland. *(continued)*

Family	**Species**	**Covariate**	**Estimate**	**Standard error**	**P**
Anatidae	Tundra Swan	wetlandsaltmasch	1.143	0.405	0.005
Anatidae	Tundra Swan	wetlandstream	-1.449	0.711	0.041
Anatidae	Tundra Swan	wetp_2km	1.918	0.021	0.000
Anatidae	Wood Duck	uplandp_2km	-0.303	0.001	0.037
Anatidae	Wood Duck	wetlandlacustrine	1.088	0.171	0.000
Anatidae	Wood Duck	wetlandlake	1.053	0.151	0.000
Anatidae	Wood Duck	wetlandpalustrine	1.338	0.141	0.000
Anatidae	Wood Duck	wetlandreservoir	1.758	0.321	0.000
Anatidae	Wood Duck	wetlandriver	1.197	0.107	0.000
Anatidae	Wood Duck	wetlandsaltmasch	1.076	0.395	0.006
Anatidae	Wood Duck	wetlandstream	0.681	0.249	0.006
Anatidae	Wood Duck	wetp_2km	0.897	0.016	0.000
Ardeidae	American Bittern	seasonality_2km	0.787	0.155	0.000
Ardeidae	American Bittern	uplandp_2km	-0.806	0.001	0.000
Ardeidae	American Bittern	wetlandephemeral	0.821	0.135	0.000
Ardeidae	American Bittern	wetlandlacustrine	0.612	0.157	0.000
Ardeidae	American Bittern	wetlandlake	0.666	0.146	0.000
Ardeidae	American Bittern	wetlandpalustrine	0.607	0.144	0.000
Ardeidae	American Bittern	wetlandpeatland	0.302	0.078	0.000
Ardeidae	American Bittern	wetlandriver	0.412	0.098	0.000
Ardeidae	American Bittern	wetlandsaltmasch	1.702	0.291	0.000
Ardeidae	American Bittern	wetp_2km	1.704	0.013	0.000
Ardeidae	Great Blue Heron	seasonality_2km	0.435	0.134	0.001
Ardeidae	Great Blue Heron	uplandp_2km	-0.676	0.001	0.000
Ardeidae	Great Blue Heron	wetlandcoastal	1.761	0.118	0.000
Ardeidae	Great Blue Heron	wetlandestuary	1.694	0.319	0.000
Ardeidae	Great Blue Heron	wetlandlacustrine	0.528	0.110	0.000
Ardeidae	Great Blue Heron	wetlandlake	0.969	0.088	0.000
Ardeidae	Great Blue Heron	wetlandother	1.486	0.101	0.000
Ardeidae	Great Blue Heron	wetlandpalustrine	0.753	0.098	0.000
Ardeidae	Great Blue Heron	wetlandpeatland	-0.210	0.068	0.002
Ardeidae	Great Blue Heron	wetlandreservoir	1.057	0.236	0.000
Ardeidae	Great Blue Heron	wetlandriver	0.569	0.071	0.000
Ardeidae	Great Blue Heron	wetlandsaltmasch	1.996	0.150	0.000
Ardeidae	Great Blue Heron	wetlandstream	0.436	0.162	0.007
Ardeidae	Great Blue Heron	wetp_2km	0.697	0.010	0.000
Ardeidae	Great Egret	uplandp_2km	-1.578	0.002	0.000
Ardeidae	Great Egret	wetlandephemeral	1.320	0.322	0.000
Ardeidae	Great Egret	wetlandestuary	2.338	0.750	0.002
Ardeidae	Great Egret	wetlandlacustrine	0.947	0.313	0.003
Ardeidae	Great Egret	wetlandlake	1.422	0.274	0.000
Ardeidae	Great Egret	wetlandother	1.623	0.332	0.000
Ardeidae	Great Egret	wetlandpalustrine	1.209	0.298	0.000
Ardeidae	Great Egret	wetlandreservoir	1.586	0.620	0.010

Table 7: Results from logistic regression of waterbird occurrence relative to wetland habitat. Results for intercept and temporal covariates are ommitted. The intercept value for wetland class was upland. *(continued)*

Family	**Species**	**Covariate**	**Estimate**	**Standard error**	**P**
Ardeidae	Great Egret	wetlandriver	0.970	0.228	0.000
Ardeidae	Great Egret	wetp_2km	1.421	0.030	0.000
Ardeidae	Green Heron	wetlandpalustrine	0.517	0.263	0.049
Ardeidae	Green Heron	wetlandreservoir	1.573	0.526	0.003
Ardeidae	Green Heron	wetlandriver	0.589	0.172	0.001
Ardeidae	Green Heron	wetp_2km	1.312	0.025	0.000
Charadriidae	Killdeer	seasonality_2km	0.476	0.040	0.000
Charadriidae	Killdeer	uplandp_2km	0.300	0.001	0.000
Charadriidae	Killdeer	wetlandephemeral	0.570	0.044	0.000
Charadriidae	Killdeer	wetlandlacustrine	-0.280	0.082	0.001
Charadriidae	Killdeer	wetlandlake	-0.225	0.072	0.002
Charadriidae	Killdeer	wetlandother	0.247	0.091	0.007
Charadriidae	Killdeer	wetlandpeatland	-0.953	0.031	0.000
Charadriidae	Killdeer	wetlandriver	0.185	0.033	0.000
Charadriidae	Killdeer	wetlandstream	-0.615	0.091	0.000
Charadriidae	Killdeer	wetp_2km	2.767	0.005	0.000
Charadriidae	Semipalmated Plover	uplandp_2km	-1.369	0.003	0.000
Charadriidae	Semipalmated Plover	wetlandcoastal	2.280	0.341	0.000
Charadriidae	Semipalmated Plover	wetlandestuary	2.113	0.763	0.006
Charadriidae	Semipalmated Plover	wetlandother	2.591	0.284	0.000
Charadriidae	Semipalmated Plover	wetlandsaltmasch	2.196	0.416	0.000
Gaviidae	Common Loon	uplandp_2km	-0.230	0.001	0.001
Gaviidae	Common Loon	wetlandcoastal	0.753	0.132	0.000
Gaviidae	Common Loon	wetlandestuary	1.099	0.294	0.000
Gaviidae	Common Loon	wetlandlacustrine	1.475	0.077	0.000
Gaviidae	Common Loon	wetlandlake	2.035	0.065	0.000
Gaviidae	Common Loon	wetlandother	1.588	0.083	0.000
Gaviidae	Common Loon	wetlandpalustrine	1.296	0.078	0.000
Gaviidae	Common Loon	wetlandpeatland	0.535	0.061	0.000
Gaviidae	Common Loon	wetlandreservoir	1.375	0.155	0.000
Gaviidae	Common Loon	wetlandriver	0.592	0.073	0.000
Gaviidae	Common Loon	wetlandsaltmasch	0.760	0.202	0.000
Gaviidae	Common Loon	wetp_2km	-3.930	0.013	0.000
Podicipedidae	Eared Grebe	seasonality_2km	3.259	1.007	0.001
Podicipedidae	Eared Grebe	uplandp_2km	3.510	0.004	0.000
Podicipedidae	Eared Grebe	wetlandcoastal	3.741	0.334	0.000
Podicipedidae	Eared Grebe	wetlandephemeral	1.446	0.241	0.000
Podicipedidae	Eared Grebe	wetlandlacustrine	3.554	0.262	0.000
Podicipedidae	Eared Grebe	wetlandlake	3.725	0.212	0.000
Podicipedidae	Eared Grebe	wetlandother	3.186	0.252	0.000
Podicipedidae	Eared Grebe	wetlandpalustrine	1.900	0.253	0.000
Podicipedidae	Eared Grebe	wetlandpeatland	-1.683	0.469	0.000
Podicipedidae	Eared Grebe	wetlandreservoir	3.927	0.539	0.000
Podicipedidae	Eared Grebe	wetlandriver	1.926	0.176	0.000

Table 7: Results from logistic regression of waterbird occurrence relative to wetland habitat. Results for intercept and temporal covariates are ommitted. The intercept value for wetland class was upland. *(continued)*

Family	**Species**	**Covariate**	**Estimate**	**Standard error**	**P**
Podicipedidae	Eared Grebe	wetp_2km	2.760	0.027	0.000
Podicipedidae	Horned Grebe	seasonality_2km	2.816	1.007	0.005
Podicipedidae	Horned Grebe	uplandp_2km	1.853	0.003	0.000
Podicipedidae	Horned Grebe	wetlandcoastal	2.383	0.364	0.000
Podicipedidae	Horned Grebe	wetlandephemeral	1.927	0.247	0.000
Podicipedidae	Horned Grebe	wetlandestuary	2.425	1.022	0.018
Podicipedidae	Horned Grebe	wetlandlacustrine	2.676	0.266	0.000
Podicipedidae	Horned Grebe	wetlandlake	2.855	0.213	0.000
Podicipedidae	Horned Grebe	wetlandother	3.566	0.207	0.000
Podicipedidae	Horned Grebe	wetlandpalustrine	1.952	0.241	0.000
Podicipedidae	Horned Grebe	wetlandreservoir	3.360	0.422	0.000
Podicipedidae	Horned Grebe	wetlandriver	1.980	0.184	0.000
Podicipedidae	Horned Grebe	wetlandsaltmasch	1.755	0.732	0.016
Podicipedidae	Horned Grebe	wetp_2km	0.937	0.025	0.000
Podicipedidae	Pied-billed Grebe	seasonality_2km	1.690	0.251	0.000
Podicipedidae	Pied-billed Grebe	uplandp_2km	0.493	0.002	0.001
Podicipedidae	Pied-billed Grebe	wetlandcoastal	1.635	0.269	0.000
Podicipedidae	Pied-billed Grebe	wetlandephemeral	1.029	0.139	0.000
Podicipedidae	Pied-billed Grebe	wetlandlacustrine	1.671	0.161	0.000
Podicipedidae	Pied-billed Grebe	wetlandlake	1.979	0.129	0.000
Podicipedidae	Pied-billed Grebe	wetlandother	1.561	0.190	0.000
Podicipedidae	Pied-billed Grebe	wetlandpalustrine	1.453	0.131	0.000
Podicipedidae	Pied-billed Grebe	wetlandriver	1.159	0.097	0.000
Podicipedidae	Pied-billed Grebe	wetlandsaltmasch	1.965	0.351	0.000
Podicipedidae	Pied-billed Grebe	wetp_2km	2.149	0.015	0.000
Podicipedidae	Red-necked Grebe	seasonality_2km	3.222	1.004	0.001
Podicipedidae	Red-necked Grebe	uplandp_2km	0.950	0.002	0.000
Podicipedidae	Red-necked Grebe	wetlandcoastal	2.232	0.263	0.000
Podicipedidae	Red-necked Grebe	wetlandephemeral	1.615	0.207	0.000
Podicipedidae	Red-necked Grebe	wetlandlacustrine	2.460	0.176	0.000
Podicipedidae	Red-necked Grebe	wetlandlake	3.044	0.142	0.000
Podicipedidae	Red-necked Grebe	wetlandother	3.226	0.155	0.000
Podicipedidae	Red-necked Grebe	wetlandpalustrine	2.274	0.154	0.000
Podicipedidae	Red-necked Grebe	wetlandreservoir	2.829	0.318	0.000
Podicipedidae	Red-necked Grebe	wetlandriver	1.695	0.136	0.000
Podicipedidae	Red-necked Grebe	wetp_2km	0.509	0.018	0.005
Rallidae	American Coot	seasonality_2km	1.883	0.189	0.000
Rallidae	American Coot	uplandp_2km	3.363	0.002	0.000
Rallidae	American Coot	wetlandcoastal	1.559	0.262	0.000
Rallidae	American Coot	wetlandephemeral	1.533	0.085	0.000
Rallidae	American Coot	wetlandlacustrine	2.206	0.136	0.000
Rallidae	American Coot	wetlandlake	2.256	0.101	0.000
Rallidae	American Coot	wetlandother	2.196	0.121	0.000
Rallidae	American Coot	wetlandpalustrine	1.615	0.095	0.000

Table 7: Results from logistic regression of waterbird occurrence relative to wetland habitat. Results for intercept and temporal covariates are ommitted. The intercept value for wetland class was upland. *(continued)*

Family	**Species**	**Covariate**	**Estimate**	**Standard error**	**P**
Rallidae	American Coot	wetlandpeatland	-1.027	0.111	0.000
Rallidae	American Coot	wetlandreservoir	2.397	0.315	0.000
Rallidae	American Coot	wetlandriver	1.644	0.064	0.000
Rallidae	American Coot	wetlandstream	-1.566	0.450	0.001
Rallidae	American Coot	wetp_2km	2.230	0.011	0.000
Rallidae	Sora	seasonality_2km	1.238	0.105	0.000
Rallidae	Sora	uplandp_2km	1.192	0.001	0.000
Rallidae	Sora	wetlandephemeral	1.121	0.073	0.000
Rallidae	Sora	wetlandlacustrine	1.189	0.132	0.000
Rallidae	Sora	wetlandlake	1.050	0.114	0.000
Rallidae	Sora	wetlandother	1.287	0.143	0.000
Rallidae	Sora	wetlandpalustrine	1.083	0.089	0.000
Rallidae	Sora	wetlandpeatland	-0.491	0.064	0.000
Rallidae	Sora	wetlandriver	0.764	0.061	0.000
Rallidae	Sora	wetlandsaltmasch	1.511	0.346	0.000
Rallidae	Sora	wetlandstream	-1.107	0.281	0.000
Rallidae	Sora	wetp_2km	2.863	0.010	0.000
Recurvirostridae	American Avocet	seasonality_2km	1.846	0.268	0.000
Recurvirostridae	American Avocet	uplandp_2km	2.686	0.003	0.000
Recurvirostridae	American Avocet	wetlandcoastal	3.095	0.268	0.000
Recurvirostridae	American Avocet	wetlandephemeral	1.824	0.122	0.000
Recurvirostridae	American Avocet	wetlandlacustrine	2.523	0.216	0.000
Recurvirostridae	American Avocet	wetlandlake	2.302	0.183	0.000
Recurvirostridae	American Avocet	wetlandother	2.644	0.188	0.000
Recurvirostridae	American Avocet	wetlandpalustrine	1.444	0.173	0.000
Recurvirostridae	American Avocet	wetlandpeatland	-2.125	0.327	0.000
Recurvirostridae	American Avocet	wetlandreservoir	2.146	0.722	0.003
Recurvirostridae	American Avocet	wetlandriver	1.733	0.106	0.000
Recurvirostridae	American Avocet	wetp_2km	3.435	0.019	0.000
Scolopacidae	American Woodcock	wetlandephemeral	-2.096	1.008	0.038
Scolopacidae	American Woodcock	wetlandpeatland	0.457	0.127	0.000
Scolopacidae	Greater Yellowlegs	seasonality_2km	0.471	0.195	0.016
Scolopacidae	Greater Yellowlegs	uplandp_2km	-0.820	0.001	0.000
Scolopacidae	Greater Yellowlegs	wetlandcoastal	1.644	0.169	0.000
Scolopacidae	Greater Yellowlegs	wetlandephemeral	1.348	0.151	0.000
Scolopacidae	Greater Yellowlegs	wetlandestuary	1.274	0.516	0.014
Scolopacidae	Greater Yellowlegs	wetlandlacustrine	0.907	0.139	0.000
Scolopacidae	Greater Yellowlegs	wetlandlake	0.723	0.130	0.000
Scolopacidae	Greater Yellowlegs	wetlandother	1.618	0.136	0.000
Scolopacidae	Greater Yellowlegs	wetlandpalustrine	1.046	0.132	0.000
Scolopacidae	Greater Yellowlegs	wetlandpeatland	0.503	0.090	0.000
Scolopacidae	Greater Yellowlegs	wetlandriver	0.805	0.101	0.000
Scolopacidae	Greater Yellowlegs	wetlandsaltmasch	2.603	0.164	0.000
Scolopacidae	Greater Yellowlegs	wetp_2km	0.319	0.014	0.019

Table 7: Results from logistic regression of waterbird occurrence relative to wetland habitat. Results for intercept and temporal covariates are ommitted. The intercept value for wetland class was upland. *(continued)*

Family	**Species**	**Covariate**	**Estimate**	**Standard error**	**P**
Scolopacidae	Least Sandpiper	uplandp_2km	-2.322	0.002	0.000
Scolopacidae	Least Sandpiper	wetlandcoastal	1.490	0.330	0.000
Scolopacidae	Least Sandpiper	wetlandephemeral	1.170	0.335	0.000
Scolopacidae	Least Sandpiper	wetlandlake	0.743	0.255	0.004
Scolopacidae	Least Sandpiper	wetlandother	1.597	0.277	0.000
Scolopacidae	Least Sandpiper	wetlandpalustrine	0.919	0.277	0.001
Scolopacidae	Least Sandpiper	wetlandpeatland	0.624	0.193	0.001
Scolopacidae	Least Sandpiper	wetlandriver	0.491	0.220	0.025
Scolopacidae	Least Sandpiper	wetlandsaltmasch	2.989	0.267	0.000
Scolopacidae	Least Sandpiper	wetp_2km	0.880	0.026	0.001
Scolopacidae	Lesser Yellowlegs	seasonality_2km	0.775	0.199	0.000
Scolopacidae	Lesser Yellowlegs	uplandp_2km	-0.780	0.001	0.000
Scolopacidae	Lesser Yellowlegs	wetlandcoastal	1.667	0.198	0.000
Scolopacidae	Lesser Yellowlegs	wetlandephemeral	1.347	0.139	0.000
Scolopacidae	Lesser Yellowlegs	wetlandlacustrine	1.148	0.149	0.000
Scolopacidae	Lesser Yellowlegs	wetlandlake	0.987	0.140	0.000
Scolopacidae	Lesser Yellowlegs	wetlandother	1.623	0.155	0.000
Scolopacidae	Lesser Yellowlegs	wetlandpalustrine	1.002	0.144	0.000
Scolopacidae	Lesser Yellowlegs	wetlandriver	0.947	0.102	0.000
Scolopacidae	Lesser Yellowlegs	wetlandsaltmasch	2.440	0.203	0.000
Scolopacidae	Lesser Yellowlegs	wetp_2km	1.503	0.014	0.000
Scolopacidae	Long-billed Curlew	uplandp_2km	4.335	0.007	0.000
Scolopacidae	Long-billed Curlew	wetlandpalustrine	-1.662	0.584	0.004
Scolopacidae	Long-billed Curlew	wetlandpeatland	-2.572	0.301	0.000
Scolopacidae	Long-billed Curlew	wetlandriver	-0.375	0.188	0.046
Scolopacidae	Long-billed Curlew	wetp_2km	1.185	0.025	0.000
Scolopacidae	Marbled Godwit	seasonality_2km	0.972	0.115	0.000
Scolopacidae	Marbled Godwit	uplandp_2km	1.070	0.002	0.000
Scolopacidae	Marbled Godwit	wetlandephemeral	0.500	0.098	0.000
Scolopacidae	Marbled Godwit	wetlandother	0.751	0.247	0.002
Scolopacidae	Marbled Godwit	wetlandpalustrine	-0.617	0.225	0.006
Scolopacidae	Marbled Godwit	wetlandpeatland	-1.774	0.133	0.000
Scolopacidae	Marbled Godwit	wetlandsaltmasch	2.204	0.362	0.000
Scolopacidae	Marbled Godwit	wetlandstream	-1.642	0.450	0.000
Scolopacidae	Marbled Godwit	wetp_2km	3.885	0.015	0.000
Scolopacidae	Solitary Sandpiper	uplandp_2km	0.586	0.002	0.014
Scolopacidae	Solitary Sandpiper	wetlandephemeral	1.022	0.272	0.000
Scolopacidae	Solitary Sandpiper	wetlandlacustrine	1.435	0.235	0.000
Scolopacidae	Solitary Sandpiper	wetlandlake	0.964	0.216	0.000
Scolopacidae	Solitary Sandpiper	wetlandpalustrine	1.118	0.214	0.000
Scolopacidae	Solitary Sandpiper	wetlandpeatland	0.327	0.147	0.026
Scolopacidae	Solitary Sandpiper	wetlandriver	0.828	0.166	0.000
Scolopacidae	Solitary Sandpiper	wetlandsaltmasch	1.580	0.477	0.001
Scolopacidae	Spotted Sandpiper	seasonality_2km	0.598	0.197	0.002

Table 7: Results from logistic regression of waterbird occurrence relative to wetland habitat. Results for intercept and temporal covariates are ommitted. The intercept value for wetland class was upland. *(continued)*

Family	**Species**	**Covariate**	**Estimate**	**Standard error**	**P**
Scolopacidae	Spotted Sandpiper	uplandp_2km	0.253	0.001	0.020
Scolopacidae	Spotted Sandpiper	wetlandcoastal	1.700	0.190	0.000
Scolopacidae	Spotted Sandpiper	wetlandephemeral	1.199	0.148	0.000
Scolopacidae	Spotted Sandpiper	wetlandestuary	1.792	0.516	0.001
Scolopacidae	Spotted Sandpiper	wetlandlacustrine	1.192	0.136	0.000
Scolopacidae	Spotted Sandpiper	wetlandlake	1.904	0.101	0.000
Scolopacidae	Spotted Sandpiper	wetlandother	2.090	0.123	0.000
Scolopacidae	Spotted Sandpiper	wetlandpalustrine	1.075	0.126	0.000
Scolopacidae	Spotted Sandpiper	wetlandpeatland	-0.526	0.104	0.000
Scolopacidae	Spotted Sandpiper	wetlandreservoir	2.819	0.182	0.000
Scolopacidae	Spotted Sandpiper	wetlandriver	1.604	0.083	0.000
Scolopacidae	Spotted Sandpiper	wetlandsaltmasch	0.897	0.390	0.022
Scolopacidae	Spotted Sandpiper	wetlandstream	0.968	0.180	0.000
Scolopacidae	Spotted Sandpiper	wetp_2km	0.284	0.013	0.029
Scolopacidae	Upland Sandpiper	seasonality_2km	0.845	0.084	0.000
Scolopacidae	Upland Sandpiper	uplandp_2km	1.468	0.002	0.000
Scolopacidae	Upland Sandpiper	wetlandephemeral	0.608	0.078	0.000
Scolopacidae	Upland Sandpiper	wetlandlake	-0.847	0.418	0.043
Scolopacidae	Upland Sandpiper	wetlandpalustrine	-0.540	0.196	0.006
Scolopacidae	Upland Sandpiper	wetlandpeatland	-1.787	0.116	0.000
Scolopacidae	Upland Sandpiper	wetlandriver	-0.182	0.089	0.041
Scolopacidae	Upland Sandpiper	wetlandstream	-1.264	0.320	0.000
Scolopacidae	Upland Sandpiper	wetp_2km	4.944	0.014	0.000
Scolopacidae	Willet	seasonality_2km	1.411	0.116	0.000
Scolopacidae	Willet	uplandp_2km	2.103	0.002	0.000
Scolopacidae	Willet	wetlandcoastal	2.483	0.200	0.000
Scolopacidae	Willet	wetlandephemeral	1.124	0.073	0.000
Scolopacidae	Willet	wetlandlacustrine	0.997	0.195	0.000
Scolopacidae	Willet	wetlandlake	0.955	0.162	0.000
Scolopacidae	Willet	wetlandother	2.044	0.135	0.000
Scolopacidae	Willet	wetlandpalustrine	0.428	0.131	0.001
Scolopacidae	Willet	wetlandpeatland	-3.276	0.246	0.000
Scolopacidae	Willet	wetlandreservoir	1.341	0.510	0.008
Scolopacidae	Willet	wetlandriver	0.675	0.069	0.000
Scolopacidae	Willet	wetlandsaltmasch	3.557	0.231	0.000
Scolopacidae	Willet	wetlandstream	-1.556	0.381	0.000
Scolopacidae	Willet	wetp_2km	3.818	0.012	0.000
Scolopacidae	Wilson's Phalarope	seasonality_2km	1.372	0.193	0.000
Scolopacidae	Wilson's Phalarope	uplandp_2km	3.613	0.003	0.000
Scolopacidae	Wilson's Phalarope	wetlandcoastal	3.243	0.309	0.000
Scolopacidae	Wilson's Phalarope	wetlandephemeral	1.541	0.119	0.000
Scolopacidae	Wilson's Phalarope	wetlandlacustrine	2.132	0.276	0.000
Scolopacidae	Wilson's Phalarope	wetlandlake	2.263	0.207	0.000
Scolopacidae	Wilson's Phalarope	wetlandother	2.455	0.213	0.000

Table 7: Results from logistic regression of waterbird occurrence relative to wetland habitat. Results for intercept and temporal covariates are ommitted. The intercept value for wetland class was upland. *(continued)*

Family	**Species**	**Covariate**	**Estimate**	**Standard error**	**P**
Scolopacidae	Wilson's Phalarope	wetlandpalustrine	1.120	0.193	0.000
Scolopacidae	Wilson's Phalarope	wetlandpeatland	-1.626	0.248	0.000
Scolopacidae	Wilson's Phalarope	wetlandriver	1.623	0.102	0.000
Scolopacidae	Wilson's Phalarope	wetp_2km	4.052	0.020	0.000
Scolopacidae	Wilson's Snipe	seasonality_2km	0.665	0.054	0.000
Scolopacidae	Wilson's Snipe	uplandp_2km	-0.385	0.001	0.000
Scolopacidae	Wilson's Snipe	wetlandcoastal	0.517	0.134	0.000
Scolopacidae	Wilson's Snipe	wetlandephemeral	0.526	0.056	0.000
Scolopacidae	Wilson's Snipe	wetlandlacustrine	0.200	0.073	0.006
Scolopacidae	Wilson's Snipe	wetlandother	0.295	0.103	0.004
Scolopacidae	Wilson's Snipe	wetlandpalustrine	0.227	0.062	0.000
Scolopacidae	Wilson's Snipe	wetlandsaltmasch	1.787	0.120	0.000
Scolopacidae	Wilson's Snipe	wetlandstream	-0.409	0.111	0.000
Scolopacidae	Wilson's Snipe	wetp_2km	2.031	0.005	0.000

Appendix B - Detection Occurrence Results

Table 8: Metrics of detection occurrence per species for ARU and point count surveys across the waterbird dataset.

Family	Species	metric	Point count	ARU	Difference
Anatidae	American Black Duck	Survey occurrence	0.007	0.000	0.007
Anatidae	American Black Duck	Site occurence	0.007	0.001	0.006
Anatidae	American Black Duck	Occurrence count	11.671	1.054	10.617
Anatidae	American Black Duck	Detection rate	0.845	0.230	0.615
Anatidae	American Wigeon	Survey occurrence	0.008	0.005	0.002
Anatidae	American Wigeon	Site occurence	0.009	0.015	-0.006
Anatidae	American Wigeon	Occurrence count	10.150	1.146	9.005
Anatidae	American Wigeon	Detection rate	0.926	0.300	0.627
Anatidae	Barnacle Goose	Survey occurrence	0.000	0.000	0.000
Anatidae	Barnacle Goose	Site occurence	0.000	0.000	0.000
Anatidae	Barnacle Goose	Occurrence count	1.221	1.000	0.221
Anatidae	Barnacle Goose	Detection rate	0.953	0.167	0.786
Anatidae	Barrow's Goldeneye	Survey occurrence	0.004	0.000	0.004
Anatidae	Barrow's Goldeneye	Site occurence	0.005	0.000	0.005
Anatidae	Barrow's Goldeneye	Occurrence count	5.454	1.071	4.383
Anatidae	Barrow's Goldeneye	Detection rate	0.953	0.549	0.404
Anatidae	Black Scoter	Survey occurrence	0.001	0.001	0.000
Anatidae	Black Scoter	Site occurence	0.001	0.003	-0.002
Anatidae	Black Scoter	Occurrence count	28.215	1.175	27.040
Anatidae	Black Scoter	Detection rate	0.811	0.237	0.574
Anatidae	Black-bellied Whistling-Duck	Survey occurrence	0.000	0.000	0.000
Anatidae	Black-bellied Whistling-Duck	Site occurence	0.000	0.000	0.000
Anatidae	Black-bellied Whistling-Duck	Occurrence count	8.019	1.000	7.019
Anatidae	Black-bellied Whistling-Duck	Detection rate	1.000	0.125	0.875
Anatidae	Blue-winged Teal	Survey occurrence	0.008	0.002	0.006
Anatidae	Blue-winged Teal	Site occurence	0.010	0.006	0.004
Anatidae	Blue-winged Teal	Occurrence count	3.085	1.314	1.771
Anatidae	Blue-winged Teal	Detection rate	0.966	0.335	0.631
Anatidae	Brant	Survey occurrence	0.000	0.000	0.000
Anatidae	Brant	Site occurence	0.000	0.000	0.000
Anatidae	Brant	Occurrence count	64.734	3.200	61.534
Anatidae	Brant	Detection rate	0.759	0.093	0.667
Anatidae	Bufflehead	Survey occurrence	0.009	0.000	0.009
Anatidae	Bufflehead	Site occurence	0.010	0.001	0.009
Anatidae	Bufflehead	Occurrence count	5.762	1.083	4.679
Anatidae	Bufflehead	Detection rate	0.893	0.165	0.729
Anatidae	Cackling Goose	Survey occurrence	0.001	0.000	0.001
Anatidae	Cackling Goose	Site occurence	0.001	0.000	0.001
Anatidae	Cackling Goose	Occurrence count	59.555	1.000	58.555
Anatidae	Cackling Goose	Detection rate	0.782	0.121	0.661
Anatidae	Canada Goose	Survey occurrence	0.054	0.042	0.012
Anatidae	Canada Goose	Site occurence	0.054	0.094	-0.040
Anatidae	Canada Goose	Occurrence count	46.012	1.887	44.125

Table 8: Metrics of detection occurrence per species for ARU and point count surveys across the waterbird dataset. *(continued)*

Family	**Species**	**metric**	**Point count**	**ARU**	**Difference**
Anatidae	Canada Goose	Detection rate	0.884	0.482	0.402
Anatidae	Canvasback	Survey occurrence	0.002	0.000	0.002
Anatidae	Canvasback	Site occurence	0.002	0.000	0.002
Anatidae	Canvasback	Occurrence count	12.561	1.000	11.561
Anatidae	Canvasback	Detection rate	0.947	0.146	0.801
Anatidae	Cinnamon Teal	Survey occurrence	0.000	0.000	0.000
Anatidae	Cinnamon Teal	Site occurence	0.001	0.000	0.001
Anatidae	Cinnamon Teal	Occurrence count	2.135	0.000	2.135
Anatidae	Cinnamon Teal	Detection rate	0.937	0.000	0.937
Anatidae	Common Eider	Survey occurrence	0.002	0.000	0.002
Anatidae	Common Eider	Site occurence	0.002	0.000	0.002
Anatidae	Common Eider	Occurrence count	29.481	2.350	27.131
Anatidae	Common Eider	Detection rate	0.887	0.184	0.703
Anatidae	Common Goldeneye	Survey occurrence	0.009	0.001	0.008
Anatidae	Common Goldeneye	Site occurence	0.009	0.002	0.006
Anatidae	Common Goldeneye	Occurrence count	9.662	1.173	8.489
Anatidae	Common Goldeneye	Detection rate	0.859	0.217	0.642
Anatidae	Common Merganser	Survey occurrence	0.008	0.001	0.007
Anatidae	Common Merganser	Site occurence	0.009	0.004	0.005
Anatidae	Common Merganser	Occurrence count	9.394	1.033	8.360
Anatidae	Common Merganser	Detection rate	0.837	0.241	0.596
Anatidae	Egyptian Goose	Survey occurrence	0.000	0.000	0.000
Anatidae	Egyptian Goose	Site occurence	0.000	0.000	0.000
Anatidae	Egyptian Goose	Occurrence count	1.000	0.000	1.000
Anatidae	Egyptian Goose	Detection rate	1.000	0.000	1.000
Anatidae	Eurasian Wigeon	Survey occurrence	0.000	0.000	0.000
Anatidae	Eurasian Wigeon	Site occurence	0.000	0.000	0.000
Anatidae	Eurasian Wigeon	Occurrence count	1.400	0.000	1.400
Anatidae	Eurasian Wigeon	Detection rate	0.821	0.000	0.821
Anatidae	Gadwall	Survey occurrence	0.007	0.006	0.000
Anatidae	Gadwall	Site occurence	0.008	0.016	-0.008
Anatidae	Gadwall	Occurrence count	3.937	1.279	2.658
Anatidae	Gadwall	Detection rate	0.958	0.402	0.556
Anatidae	Garganey	Survey occurrence	0.000	0.000	0.000
Anatidae	Garganey	Site occurence	0.000	0.000	0.000
Anatidae	Garganey	Occurrence count	1.000	0.000	1.000
Anatidae	Garganey	Detection rate	1.000	0.000	1.000
Anatidae	Graylag Goose	Survey occurrence	0.000	0.000	0.000
Anatidae	Graylag Goose	Site occurence	0.000	0.000	0.000
Anatidae	Graylag Goose	Occurrence count	1.933	0.000	1.933
Anatidae	Graylag Goose	Detection rate	0.933	0.000	0.933
Anatidae	Greater Scaup	Survey occurrence	0.002	0.000	0.002
Anatidae	Greater Scaup	Site occurence	0.002	0.000	0.002
Anatidae	Greater Scaup	Occurrence count	28.505	1.000	27.505
Anatidae	Greater Scaup	Detection rate	0.843	0.362	0.480
Anatidae	Greater White-fronted Goose	Survey occurrence	0.001	0.001	0.000

Table 8: Metrics of detection occurrence per species for ARU and point count surveys across the waterbird dataset. *(continued)*

Family	**Species**	**metric**	**Point count**	**ARU**	**Difference**
Anatidae	Greater White-fronted Goose	Site occurence	0.001	0.001	0.000
Anatidae	Greater White-fronted Goose	Occurrence count	75.850	1.770	74.080
Anatidae	Greater White-fronted Goose	Detection rate	0.870	0.277	0.593
Anatidae	Green-winged Teal	Survey occurrence	0.007	0.005	0.002
Anatidae	Green-winged Teal	Site occurence	0.008	0.015	-0.006
Anatidae	Green-winged Teal	Occurrence count	6.158	1.157	5.001
Anatidae	Green-winged Teal	Detection rate	0.914	0.243	0.672
Anatidae	Harlequin Duck	Survey occurrence	0.001	0.000	0.001
Anatidae	Harlequin Duck	Site occurence	0.001	0.000	0.001
Anatidae	Harlequin Duck	Occurrence count	5.326	1.500	3.826
Anatidae	Harlequin Duck	Detection rate	0.873	1.000	-0.127
Anatidae	Hawaiian Duck	Survey occurrence	0.000	0.000	0.000
Anatidae	Hawaiian Duck	Site occurence	0.000	0.000	0.000
Anatidae	Hawaiian Duck	Occurrence count	1.000	0.000	1.000
Anatidae	Hawaiian Duck	Detection rate	0.306	0.000	0.306
Anatidae	Hooded Merganser	Survey occurrence	0.005	0.000	0.005
Anatidae	Hooded Merganser	Site occurence	0.005	0.001	0.004
Anatidae	Hooded Merganser	Occurrence count	4.683	1.014	3.670
Anatidae	Hooded Merganser	Detection rate	0.804	0.152	0.651
Anatidae	King Eider	Survey occurrence	0.000	0.000	0.000
Anatidae	King Eider	Site occurence	0.000	0.000	0.000
Anatidae	King Eider	Occurrence count	8.656	0.000	8.656
Anatidae	King Eider	Detection rate	0.869	0.000	0.869
Anatidae	Lesser Scaup	Survey occurrence	0.005	0.000	0.004
Anatidae	Lesser Scaup	Site occurence	0.006	0.002	0.004
Anatidae	Lesser Scaup	Occurrence count	7.463	1.135	6.328
Anatidae	Lesser Scaup	Detection rate	0.944	0.243	0.701
Anatidae	Lesser Snow Goose White-morph	Survey occurrence	0.000	0.000	0.000
Anatidae	Lesser Snow Goose White-morph	Site occurence	0.000	0.000	0.000
Anatidae	Lesser Snow Goose White-morph	Occurrence count	1.000	0.000	1.000
Anatidae	Lesser Snow Goose White-morph	Detection rate	1.000	0.000	1.000
Anatidae	Long-tailed Duck	Survey occurrence	0.002	0.004	-0.002
Anatidae	Long-tailed Duck	Site occurence	0.002	0.006	-0.004
Anatidae	Long-tailed Duck	Occurrence count	18.886	1.167	17.719
Anatidae	Long-tailed Duck	Detection rate	0.863	0.390	0.473
Anatidae	Mallard	Survey occurrence	0.050	0.010	0.041
Anatidae	Mallard	Site occurence	0.054	0.030	0.024
Anatidae	Mallard	Occurrence count	10.485	1.132	9.354
Anatidae	Mallard	Detection rate	0.925	0.347	0.578
Anatidae	Mandarin Duck	Survey occurrence	0.000	0.000	0.000
Anatidae	Mandarin Duck	Site occurence	0.000	0.000	0.000
Anatidae	Mandarin Duck	Occurrence count	1.200	0.000	1.200
Anatidae	Mandarin Duck	Detection rate	1.000	0.000	1.000
Anatidae	Muscovy Duck	Survey occurrence	0.000	0.000	0.000
Anatidae	Muscovy Duck	Site occurence	0.000	0.000	0.000
Anatidae	Muscovy Duck	Occurrence count	2.102	1.000	1.102

Table 8: Metrics of detection occurrence per species for ARU and point count surveys across the waterbird dataset. *(continued)*

Family	**Species**	**metric**	**Point count**	**ARU**	**Difference**
Anatidae	Muscovy Duck	Detection rate	0.896	1.000	-0.104
Anatidae	Mute Swan	Survey occurrence	0.002	0.000	0.001
Anatidae	Mute Swan	Site occurence	0.002	0.001	0.000
Anatidae	Mute Swan	Occurrence count	7.931	1.024	6.908
Anatidae	Mute Swan	Detection rate	0.891	0.148	0.743
Anatidae	Northern Pintail	Survey occurrence	0.006	0.003	0.003
Anatidae	Northern Pintail	Site occurence	0.007	0.009	-0.002
Anatidae	Northern Pintail	Occurrence count	11.049	1.108	9.941
Anatidae	Northern Pintail	Detection rate	0.943	0.299	0.644
Anatidae	Northern Shoveler	Survey occurrence	0.008	0.003	0.005
Anatidae	Northern Shoveler	Site occurence	0.009	0.008	0.001
Anatidae	Northern Shoveler	Occurrence count	4.355	1.213	3.142
Anatidae	Northern Shoveler	Detection rate	0.965	0.343	0.621
Anatidae	Pink-footed Goose	Survey occurrence	0.000	0.000	0.000
Anatidae	Pink-footed Goose	Site occurence	0.000	0.000	0.000
Anatidae	Pink-footed Goose	Occurrence count	1.371	0.000	1.371
Anatidae	Pink-footed Goose	Detection rate	0.951	0.000	0.951
Anatidae	Red-breasted Merganser	Survey occurrence	0.003	0.000	0.003
Anatidae	Red-breasted Merganser	Site occurence	0.003	0.000	0.003
Anatidae	Red-breasted Merganser	Occurrence count	9.922	1.000	8.922
Anatidae	Red-breasted Merganser	Detection rate	0.829	0.278	0.551
Anatidae	Redhead	Survey occurrence	0.003	0.001	0.002
Anatidae	Redhead	Site occurence	0.003	0.002	0.001
Anatidae	Redhead	Occurrence count	23.026	1.115	21.911
Anatidae	Redhead	Detection rate	0.932	0.258	0.675
Anatidae	Ring-necked Duck	Survey occurrence	0.006	0.001	0.005
Anatidae	Ring-necked Duck	Site occurence	0.007	0.003	0.003
Anatidae	Ring-necked Duck	Occurrence count	8.525	1.140	7.385
Anatidae	Ring-necked Duck	Detection rate	0.898	0.209	0.688
Anatidae	Ross's Goose	Survey occurrence	0.000	0.000	0.000
Anatidae	Ross's Goose	Site occurence	0.000	0.000	0.000
Anatidae	Ross's Goose	Occurrence count	7.397	0.000	7.397
Anatidae	Ross's Goose	Detection rate	0.883	0.000	0.883
Anatidae	Ruddy Duck	Survey occurrence	0.002	0.001	0.001
Anatidae	Ruddy Duck	Site occurence	0.002	0.003	0.000
Anatidae	Ruddy Duck	Occurrence count	4.142	1.098	3.044
Anatidae	Ruddy Duck	Detection rate	0.959	0.292	0.667
Anatidae	Smew	Survey occurrence	0.000	0.000	0.000
Anatidae	Smew	Site occurence	0.000	0.000	0.000
Anatidae	Smew	Occurrence count	1.000	0.000	1.000
Anatidae	Smew	Detection rate	1.000	0.000	1.000
Anatidae	Snow Goose	Survey occurrence	0.003	0.001	0.002
Anatidae	Snow Goose	Site occurence	0.003	0.001	0.002
Anatidae	Snow Goose	Occurrence count	651.957	3.515	100.000
Anatidae	Snow Goose	Detection rate	0.813	0.090	0.723

Table 8: Metrics of detection occurrence per species for ARU and point count surveys across the waterbird dataset. *(continued)*

Family	**Species**	**metric**	**Point count**	**ARU**	**Difference**
Anatidae	Surf Scoter	Survey occurrence	0.001	0.000	0.001
Anatidae	Surf Scoter	Site occurence	0.001	0.002	0.000
Anatidae	Surf Scoter	Occurrence count	20.600	1.072	19.528
Anatidae	Surf Scoter	Detection rate	0.843	0.386	0.456
Anatidae	Taiga Bean-Goose	Survey occurrence	0.000	0.000	0.000
Anatidae	Taiga Bean-Goose	Site occurence	0.000	0.000	0.000
Anatidae	Taiga Bean-Goose	Occurrence count	1.000	0.000	1.000
Anatidae	Taiga Bean-Goose	Detection rate	1.000	0.000	1.000
Anatidae	Trumpeter Swan	Survey occurrence	0.003	0.002	0.002
Anatidae	Trumpeter Swan	Site occurence	0.004	0.004	0.000
Anatidae	Trumpeter Swan	Occurrence count	9.633	1.184	8.450
Anatidae	Trumpeter Swan	Detection rate	0.876	0.279	0.597
Anatidae	Tufted Duck	Survey occurrence	0.000	0.000	0.000
Anatidae	Tufted Duck	Site occurence	0.000	0.000	0.000
Anatidae	Tufted Duck	Occurrence count	1.086	0.000	1.086
Anatidae	Tufted Duck	Detection rate	0.880	0.000	0.880
Anatidae	Tundra Bean-Goose	Survey occurrence	0.000	0.000	0.000
Anatidae	Tundra Bean-Goose	Site occurence	0.000	0.000	0.000
Anatidae	Tundra Bean-Goose	Occurrence count	1.000	0.000	1.000
Anatidae	Tundra Bean-Goose	Detection rate	1.000	0.000	1.000
Anatidae	Tundra Swan	Survey occurrence	0.002	0.001	0.000
Anatidae	Tundra Swan	Site occurence	0.002	0.003	-0.001
Anatidae	Tundra Swan	Occurrence count	52.768	1.696	51.072
Anatidae	Tundra Swan	Detection rate	0.866	0.374	0.492
Anatidae	White-winged Scoter	Survey occurrence	0.001	0.000	0.001
Anatidae	White-winged Scoter	Site occurence	0.001	0.000	0.001
Anatidae	White-winged Scoter	Occurrence count	12.671	1.000	11.671
Anatidae	White-winged Scoter	Detection rate	0.880	0.143	0.737
Anatidae	Whooper Swan	Survey occurrence	0.000	0.000	0.000
Anatidae	Whooper Swan	Site occurence	0.000	0.000	0.000
Anatidae	Whooper Swan	Occurrence count	1.000	0.000	1.000
Anatidae	Whooper Swan	Detection rate	1.000	0.000	1.000
Anatidae	Wood Duck	Survey occurrence	0.005	0.000	0.005
Anatidae	Wood Duck	Site occurence	0.005	0.002	0.004
Anatidae	Wood Duck	Occurrence count	3.386	1.056	2.330
Anatidae	Wood Duck	Detection rate	0.854	0.224	0.630
Ardeidae	American Bittern	Survey occurrence	0.005	0.011	-0.005
Ardeidae	American Bittern	Site occurence	0.007	0.027	-0.020
Ardeidae	American Bittern	Occurrence count	1.067	1.112	-0.045
Ardeidae	American Bittern	Detection rate	0.957	0.364	0.593
Ardeidae	Black-crowned Night Heron	Survey occurrence	0.001	0.000	0.000
Ardeidae	Black-crowned Night Heron	Site occurence	0.001	0.001	0.000
Ardeidae	Black-crowned Night Heron	Occurrence count	1.864	1.000	0.864
Ardeidae	Black-crowned Night Heron	Detection rate	0.852	0.190	0.662
Ardeidae	Chinese Pond-Heron	Survey occurrence	0.000	0.000	0.000

Table 8: Metrics of detection occurrence per species for ARU and point count surveys across the waterbird dataset. *(continued)*

Family	**Species**	**metric**	**Point count**	**ARU**	**Difference**
Ardeidae	Chinese Pond-Heron	Site occurence	0.000	0.000	0.000
Ardeidae	Chinese Pond-Heron	Occurrence count	1.000	0.000	1.000
Ardeidae	Chinese Pond-Heron	Detection rate	0.500	0.000	0.500
Ardeidae	Gray Heron	Survey occurrence	0.000	0.000	0.000
Ardeidae	Gray Heron	Site occurence	0.000	0.000	0.000
Ardeidae	Gray Heron	Occurrence count	1.000	0.000	1.000
Ardeidae	Gray Heron	Detection rate	0.964	0.000	0.964
Ardeidae	Great Blue Heron	Survey occurrence	0.014	0.001	0.012
Ardeidae	Great Blue Heron	Site occurence	0.016	0.005	0.011
Ardeidae	Great Blue Heron	Occurrence count	1.737	1.080	0.657
Ardeidae	Great Blue Heron	Detection rate	0.877	0.215	0.662
Ardeidae	Great Egret	Survey occurrence	0.001	0.000	0.001
Ardeidae	Great Egret	Site occurence	0.002	0.000	0.002
Ardeidae	Great Egret	Occurrence count	2.711	2.000	0.711
Ardeidae	Great Egret	Detection rate	0.875	0.086	0.790
Ardeidae	Green Heron	Survey occurrence	0.002	0.000	0.002
Ardeidae	Green Heron	Site occurence	0.002	0.001	0.001
Ardeidae	Green Heron	Occurrence count	1.211	1.077	0.135
Ardeidae	Green Heron	Detection rate	0.908	0.469	0.439
Ardeidae	Least Bittern	Survey occurrence	0.000	0.001	-0.001
Ardeidae	Least Bittern	Site occurence	0.000	0.002	-0.002
Ardeidae	Least Bittern	Occurrence count	1.174	1.135	0.039
Ardeidae	Least Bittern	Detection rate	0.865	0.140	0.725
Ardeidae	Little Blue Heron	Survey occurrence	0.000	0.000	0.000
Ardeidae	Little Blue Heron	Site occurence	0.000	0.000	0.000
Ardeidae	Little Blue Heron	Occurrence count	1.916	0.000	1.916
Ardeidae	Little Blue Heron	Detection rate	0.978	0.000	0.978
Ardeidae	Little Egret	Survey occurrence	0.000	0.000	0.000
Ardeidae	Little Egret	Site occurence	0.000	0.000	0.000
Ardeidae	Little Egret	Occurrence count	1.000	0.000	1.000
Ardeidae	Little Egret	Detection rate	0.800	0.000	0.800
Ardeidae	Reddish Egret	Survey occurrence	0.000	0.000	0.000
Ardeidae	Reddish Egret	Site occurence	0.000	0.000	0.000
Ardeidae	Reddish Egret	Occurrence count	1.000	0.000	1.000
Ardeidae	Reddish Egret	Detection rate	1.000	0.000	1.000
Ardeidae	Snowy Egret	Survey occurrence	0.000	0.000	0.000
Ardeidae	Snowy Egret	Site occurence	0.000	0.000	0.000
Ardeidae	Snowy Egret	Occurrence count	3.279	0.000	3.279
Ardeidae	Snowy Egret	Detection rate	0.961	0.000	0.961
Ardeidae	Tricolored Heron	Survey occurrence	0.000	0.000	0.000
Ardeidae	Tricolored Heron	Site occurence	0.000	0.000	0.000
Ardeidae	Tricolored Heron	Occurrence count	1.000	0.000	1.000
Ardeidae	Tricolored Heron	Detection rate	1.000	0.000	1.000
Ardeidae	Yellow-crowned Night Heron	Survey occurrence	0.000	0.000	0.000
Ardeidae	Yellow-crowned Night Heron	Site occurence	0.000	0.000	0.000

Table 8: Metrics of detection occurrence per species for ARU and point count surveys across the waterbird dataset. *(continued)*

Family	**Species**	**metric**	**Point count**	**ARU**	**Difference**
Ardeidae	Yellow-crowned Night Heron	Occurrence count	1.095	0.000	1.095
Ardeidae	Yellow-crowned Night Heron	Detection rate	0.957	0.000	0.957
Charadriidae	American Golden-Plover	Survey occurrence	0.001	0.000	0.000
Charadriidae	American Golden-Plover	Site occurence	0.001	0.002	-0.001
Charadriidae	American Golden-Plover	Occurrence count	8.971	1.079	7.892
Charadriidae	American Golden-Plover	Detection rate	0.936	0.252	0.684
Charadriidae	Black-bellied Plover	Survey occurrence	0.001	0.000	0.001
Charadriidae	Black-bellied Plover	Site occurence	0.001	0.001	0.000
Charadriidae	Black-bellied Plover	Occurrence count	23.321	1.000	22.321
Charadriidae	Black-bellied Plover	Detection rate	0.840	0.219	0.621
Charadriidae	Common Ringed Plover	Survey occurrence	0.000	0.000	0.000
Charadriidae	Common Ringed Plover	Site occurence	0.000	0.000	0.000
Charadriidae	Common Ringed Plover	Occurrence count	1.000	0.000	1.000
Charadriidae	Common Ringed Plover	Detection rate	0.750	0.000	0.750
Charadriidae	European Golden-Plover	Survey occurrence	0.000	0.000	0.000
Charadriidae	European Golden-Plover	Site occurence	0.000	0.000	0.000
Charadriidae	European Golden-Plover	Occurrence count	1.000	0.000	1.000
Charadriidae	European Golden-Plover	Detection rate	1.000	0.000	1.000
Charadriidae	Killdeer	Survey occurrence	0.048	0.008	0.040
Charadriidae	Killdeer	Site occurence	0.058	0.026	0.032
Charadriidae	Killdeer	Occurrence count	1.740	1.144	0.596
Charadriidae	Killdeer	Detection rate	0.968	0.500	0.469
Charadriidae	Northern Lapwing	Survey occurrence	0.000	0.000	0.000
Charadriidae	Northern Lapwing	Site occurence	0.000	0.000	0.000
Charadriidae	Northern Lapwing	Occurrence count	1.000	0.000	1.000
Charadriidae	Northern Lapwing	Detection rate	1.000	0.000	1.000
Charadriidae	Pacific Golden-Plover	Survey occurrence	0.000	0.000	0.000
Charadriidae	Pacific Golden-Plover	Site occurence	0.000	0.000	0.000
Charadriidae	Pacific Golden-Plover	Occurrence count	1.540	0.000	1.540
Charadriidae	Pacific Golden-Plover	Detection rate	0.992	0.000	0.992
Charadriidae	Piping Plover	Survey occurrence	0.000	0.000	0.000
Charadriidae	Piping Plover	Site occurence	0.000	0.000	0.000
Charadriidae	Piping Plover	Occurrence count	1.958	0.000	1.958
Charadriidae	Piping Plover	Detection rate	0.883	0.000	0.883
Charadriidae	Semipalmated Plover	Survey occurrence	0.001	0.001	0.000
Charadriidae	Semipalmated Plover	Site occurence	0.001	0.002	0.000
Charadriidae	Semipalmated Plover	Occurrence count	6.877	1.046	5.831
Charadriidae	Semipalmated Plover	Detection rate	0.850	0.145	0.705
Gaviidae	Common Loon	Survey occurrence	0.017	0.029	-0.013
Gaviidae	Common Loon	Site occurence	0.018	0.078	-0.060
Gaviidae	Common Loon	Occurrence count	1.501	1.151	0.350
Gaviidae	Common Loon	Detection rate	0.902	0.366	0.536
Gaviidae	Pacific Loon	Survey occurrence	0.001	0.001	0.000
Gaviidae	Pacific Loon	Site occurence	0.001	0.004	-0.003
Gaviidae	Pacific Loon	Occurrence count	3.527	1.144	2.382

Table 8: Metrics of detection occurrence per species for ARU and point count surveys across the waterbird dataset. *(continued)*

Family	**Species**	**metric**	**Point count**	**ARU**	**Difference**
Gaviidae	Pacific Loon	Detection rate	0.953	0.349	0.604
Gaviidae	Red-throated Loon	Survey occurrence	0.001	0.003	-0.003
Gaviidae	Red-throated Loon	Site occurence	0.001	0.007	-0.006
Gaviidae	Red-throated Loon	Occurrence count	2.141	1.291	0.850
Gaviidae	Red-throated Loon	Detection rate	0.885	0.355	0.530
Gaviidae	Yellow-billed Loon	Survey occurrence	0.000	0.000	0.000
Gaviidae	Yellow-billed Loon	Site occurence	0.000	0.000	0.000
Gaviidae	Yellow-billed Loon	Occurrence count	1.241	1.273	-0.031
Gaviidae	Yellow-billed Loon	Detection rate	0.983	0.172	0.811
Haematopodidae	American Oystercatcher	Survey occurrence	0.000	0.000	0.000
Haematopodidae	American Oystercatcher	Site occurence	0.000	0.000	0.000
Haematopodidae	American Oystercatcher	Occurrence count	1.917	0.000	1.917
Haematopodidae	American Oystercatcher	Detection rate	0.718	0.000	0.718
Haematopodidae	Black Oystercatcher	Survey occurrence	0.000	0.000	0.000
Haematopodidae	Black Oystercatcher	Site occurence	0.000	0.000	0.000
Haematopodidae	Black Oystercatcher	Occurrence count	4.421	0.000	4.421
Haematopodidae	Black Oystercatcher	Detection rate	0.831	0.000	0.831
Haematopodidae	Eurasian Oystercatcher	Survey occurrence	0.000	0.000	0.000
Haematopodidae	Eurasian Oystercatcher	Site occurence	0.000	0.000	0.000
Haematopodidae	Eurasian Oystercatcher	Occurrence count	1.000	0.000	1.000
Haematopodidae	Eurasian Oystercatcher	Detection rate	1.000	0.000	1.000
Podicipedidae	Clark's Grebe	Survey occurrence	0.000	0.000	0.000
Podicipedidae	Clark's Grebe	Site occurence	0.000	0.000	0.000
Podicipedidae	Clark's Grebe	Occurrence count	1.000	1.000	0.000
Podicipedidae	Clark's Grebe	Detection rate	1.000	0.048	0.952
Podicipedidae	Eared Grebe	Survey occurrence	0.001	0.001	-0.001
Podicipedidae	Eared Grebe	Site occurence	0.001	0.005	-0.004
Podicipedidae	Eared Grebe	Occurrence count	9.175	1.798	7.377
Podicipedidae	Eared Grebe	Detection rate	0.958	0.315	0.643
Podicipedidae	Horned Grebe	Survey occurrence	0.003	0.000	0.002
Podicipedidae	Horned Grebe	Site occurence	0.003	0.001	0.002
Podicipedidae	Horned Grebe	Occurrence count	2.803	1.039	1.764
Podicipedidae	Horned Grebe	Detection rate	0.923	0.153	0.770
Podicipedidae	Pied-billed Grebe	Survey occurrence	0.004	0.009	-0.005
Podicipedidae	Pied-billed Grebe	Site occurence	0.005	0.026	-0.022
Podicipedidae	Pied-billed Grebe	Occurrence count	1.390	1.129	0.261
Podicipedidae	Pied-billed Grebe	Detection rate	0.915	0.427	0.488
Podicipedidae	Red-necked Grebe	Survey occurrence	0.004	0.009	-0.006
Podicipedidae	Red-necked Grebe	Site occurence	0.004	0.025	-0.021
Podicipedidae	Red-necked Grebe	Occurrence count	2.809	1.558	1.252
Podicipedidae	Red-necked Grebe	Detection rate	0.904	0.505	0.399
Podicipedidae	Western Grebe	Survey occurrence	0.000	0.001	0.000
Podicipedidae	Western Grebe	Site occurence	0.000	0.001	-0.001
Podicipedidae	Western Grebe	Occurrence count	10.535	1.557	8.978
Podicipedidae	Western Grebe	Detection rate	0.863	0.341	0.522

Table 8: Metrics of detection occurrence per species for ARU and point count surveys across the waterbird dataset. *(continued)*

Family	**Species**	**metric**	**Point count**	**ARU**	**Difference**
Rallidae	American Coot	Survey occurrence	0.006	0.011	-0.005
Rallidae	American Coot	Site occurence	0.008	0.030	-0.023
Rallidae	American Coot	Occurrence count	9.264	1.184	8.080
Rallidae	American Coot	Detection rate	0.966	0.473	0.493
Rallidae	Clapper Rail	Survey occurrence	0.000	0.000	0.000
Rallidae	Clapper Rail	Site occurence	0.000	0.000	0.000
Rallidae	Clapper Rail	Occurrence count	1.986	0.000	1.986
Rallidae	Clapper Rail	Detection rate	1.000	0.000	1.000
Rallidae	Common Gallinule	Survey occurrence	0.000	0.001	0.000
Rallidae	Common Gallinule	Site occurence	0.000	0.001	-0.001
Rallidae	Common Gallinule	Occurrence count	1.672	1.429	0.244
Rallidae	Common Gallinule	Detection rate	0.835	0.192	0.644
Rallidae	King Rail	Survey occurrence	0.000	0.000	0.000
Rallidae	King Rail	Site occurence	0.000	0.000	0.000
Rallidae	King Rail	Occurrence count	1.000	1.000	0.000
Rallidae	King Rail	Detection rate	1.000	0.029	0.971
Rallidae	Sora	Survey occurrence	0.009	0.033	-0.024
Rallidae	Sora	Site occurence	0.011	0.072	-0.061
Rallidae	Sora	Occurrence count	1.210	1.576	-0.367
Rallidae	Sora	Detection rate	0.973	0.548	0.425
Rallidae	Virginia Rail	Survey occurrence	0.001	0.003	-0.002
Rallidae	Virginia Rail	Site occurence	0.001	0.007	-0.006
Rallidae	Virginia Rail	Occurrence count	1.289	1.143	0.146
Rallidae	Virginia Rail	Detection rate	0.899	0.257	0.642
Rallidae	Yellow Rail	Survey occurrence	0.000	0.006	-0.006
Rallidae	Yellow Rail	Site occurence	0.000	0.007	-0.007
Rallidae	Yellow Rail	Occurrence count	1.433	1.673	-0.240
Rallidae	Yellow Rail	Detection rate	0.969	0.567	0.402
Recurvirostridae	American Avocet	Survey occurrence	0.001	0.001	0.001
Recurvirostridae	American Avocet	Site occurence	0.002	0.003	-0.001
Recurvirostridae	American Avocet	Occurrence count	3.816	1.308	2.508
Recurvirostridae	American Avocet	Detection rate	0.977	0.294	0.682
Recurvirostridae	Black-necked Stilt	Survey occurrence	0.000	0.000	0.000
Recurvirostridae	Black-necked Stilt	Site occurence	0.000	0.000	0.000
Recurvirostridae	Black-necked Stilt	Occurrence count	3.441	1.000	2.441
Recurvirostridae	Black-necked Stilt	Detection rate	0.962	1.000	-0.038
Scolopacidae	American Woodcock	Survey occurrence	0.003	0.001	0.002
Scolopacidae	American Woodcock	Site occurence	0.002	0.001	0.001
Scolopacidae	American Woodcock	Occurrence count	1.474	1.345	0.129
Scolopacidae	American Woodcock	Detection rate	0.772	0.100	0.672
Scolopacidae	Baird's Sandpiper	Survey occurrence	0.000	0.000	0.000
Scolopacidae	Baird's Sandpiper	Site occurence	0.000	0.000	0.000
Scolopacidae	Baird's Sandpiper	Occurrence count	5.259	1.500	3.759
Scolopacidae	Baird's Sandpiper	Detection rate	0.837	0.348	0.489
Scolopacidae	Bar-tailed Godwit	Survey occurrence	0.000	0.000	0.000

Table 8: Metrics of detection occurrence per species for ARU and point count surveys across the waterbird dataset. *(continued)*

Family	**Species**	**metric**	**Point count**	**ARU**	**Difference**
Scolopacidae	Bar-tailed Godwit	Site occurence	0.000	0.000	0.000
Scolopacidae	Bar-tailed Godwit	Occurrence count	1.200	0.000	1.200
Scolopacidae	Bar-tailed Godwit	Detection rate	0.872	0.000	0.872
Scolopacidae	Black Turnstone	Survey occurrence	0.000	0.000	0.000
Scolopacidae	Black Turnstone	Site occurence	0.000	0.000	0.000
Scolopacidae	Black Turnstone	Occurrence count	16.594	0.000	16.594
Scolopacidae	Black Turnstone	Detection rate	0.815	0.000	0.815
Scolopacidae	Black-tailed Godwit	Survey occurrence	0.000	0.000	0.000
Scolopacidae	Black-tailed Godwit	Site occurence	0.000	0.000	0.000
Scolopacidae	Black-tailed Godwit	Occurrence count	1.333	0.000	1.333
Scolopacidae	Black-tailed Godwit	Detection rate	1.000	0.000	1.000
Scolopacidae	Bristle-thighed Curlew	Survey occurrence	0.000	0.000	0.000
Scolopacidae	Bristle-thighed Curlew	Site occurence	0.000	0.000	0.000
Scolopacidae	Bristle-thighed Curlew	Occurrence count	1.000	0.000	1.000
Scolopacidae	Bristle-thighed Curlew	Detection rate	1.000	0.000	1.000
Scolopacidae	Buff-breasted Sandpiper	Survey occurrence	0.000	0.000	0.000
Scolopacidae	Buff-breasted Sandpiper	Site occurence	0.000	0.000	0.000
Scolopacidae	Buff-breasted Sandpiper	Occurrence count	3.658	0.000	3.658
Scolopacidae	Buff-breasted Sandpiper	Detection rate	0.954	0.000	0.954
Scolopacidae	Common Sandpiper	Survey occurrence	0.000	0.000	0.000
Scolopacidae	Common Sandpiper	Site occurence	0.000	0.000	0.000
Scolopacidae	Common Sandpiper	Occurrence count	0.000	0.000	0.000
Scolopacidae	Common Sandpiper	Detection rate	0.000	0.000	0.000
Scolopacidae	Common Snipe	Survey occurrence	0.001	0.000	0.001
Scolopacidae	Common Snipe	Site occurence	0.001	0.000	0.001
Scolopacidae	Common Snipe	Occurrence count	1.139	1.000	0.139
Scolopacidae	Common Snipe	Detection rate	0.682	0.697	-0.015
Scolopacidae	Curlew Sandpiper	Survey occurrence	0.000	0.000	0.000
Scolopacidae	Curlew Sandpiper	Site occurence	0.000	0.000	0.000
Scolopacidae	Curlew Sandpiper	Occurrence count	1.000	0.000	1.000
Scolopacidae	Curlew Sandpiper	Detection rate	1.000	0.000	1.000
Scolopacidae	Dunlin	Survey occurrence	0.001	0.000	0.000
Scolopacidae	Dunlin	Site occurence	0.001	0.000	0.000
Scolopacidae	Dunlin	Occurrence count	23.972	1.136	22.836
Scolopacidae	Dunlin	Detection rate	0.870	0.299	0.571
Scolopacidae	Greater Yellowlegs	Survey occurrence	0.005	0.014	-0.009
Scolopacidae	Greater Yellowlegs	Site occurence	0.006	0.025	-0.020
Scolopacidae	Greater Yellowlegs	Occurrence count	2.859	1.136	1.723
Scolopacidae	Greater Yellowlegs	Detection rate	0.888	0.389	0.499
Scolopacidae	Hudsonian Godwit	Survey occurrence	0.000	0.001	0.000
Scolopacidae	Hudsonian Godwit	Site occurence	0.000	0.001	-0.001
Scolopacidae	Hudsonian Godwit	Occurrence count	2.620	1.074	1.545
Scolopacidae	Hudsonian Godwit	Detection rate	0.930	0.188	0.742
Scolopacidae	Least Sandpiper	Survey occurrence	0.002	0.007	-0.005
Scolopacidae	Least Sandpiper	Site occurence	0.002	0.010	-0.008

Table 8: Metrics of detection occurrence per species for ARU and point count surveys across the waterbird dataset. *(continued)*

Family	**Species**	**metric**	**Point count**	**ARU**	**Difference**
Scolopacidae	Least Sandpiper	Occurrence count	4.399	1.086	3.313
Scolopacidae	Least Sandpiper	Detection rate	0.865	0.269	0.596
Scolopacidae	Lesser Yellowlegs	Survey occurrence	0.005	0.018	-0.012
Scolopacidae	Lesser Yellowlegs	Site occurence	0.006	0.029	-0.022
Scolopacidae	Lesser Yellowlegs	Occurrence count	3.063	1.178	1.885
Scolopacidae	Lesser Yellowlegs	Detection rate	0.936	0.435	0.501
Scolopacidae	Long-billed Curlew	Survey occurrence	0.002	0.002	0.000
Scolopacidae	Long-billed Curlew	Site occurence	0.003	0.008	-0.005
Scolopacidae	Long-billed Curlew	Occurrence count	1.373	1.177	0.196
Scolopacidae	Long-billed Curlew	Detection rate	0.993	0.453	0.539
Scolopacidae	Long-billed Dowitcher	Survey occurrence	0.000	0.000	0.000
Scolopacidae	Long-billed Dowitcher	Site occurence	0.000	0.000	0.000
Scolopacidae	Long-billed Dowitcher	Occurrence count	17.723	1.057	16.666
Scolopacidae	Long-billed Dowitcher	Detection rate	0.891	0.061	0.830
Scolopacidae	Marbled Godwit	Survey occurrence	0.005	0.006	-0.001
Scolopacidae	Marbled Godwit	Site occurence	0.006	0.019	-0.013
Scolopacidae	Marbled Godwit	Occurrence count	1.497	1.214	0.283
Scolopacidae	Marbled Godwit	Detection rate	0.992	0.440	0.553
Scolopacidae	Marsh Sandpiper	Survey occurrence	0.000	0.000	0.000
Scolopacidae	Marsh Sandpiper	Site occurence	0.000	0.000	0.000
Scolopacidae	Marsh Sandpiper	Occurrence count	1.000	0.000	1.000
Scolopacidae	Marsh Sandpiper	Detection rate	1.000	0.000	1.000
Scolopacidae	Pectoral Sandpiper	Survey occurrence	0.000	0.000	0.000
Scolopacidae	Pectoral Sandpiper	Site occurence	0.000	0.000	0.000
Scolopacidae	Pectoral Sandpiper	Occurrence count	7.625	1.050	6.575
Scolopacidae	Pectoral Sandpiper	Detection rate	0.853	0.128	0.725
Scolopacidae	Purple Sandpiper	Survey occurrence	0.000	0.000	0.000
Scolopacidae	Purple Sandpiper	Site occurence	0.000	0.000	0.000
Scolopacidae	Purple Sandpiper	Occurrence count	16.797	0.000	16.797
Scolopacidae	Purple Sandpiper	Detection rate	0.846	0.000	0.846
Scolopacidae	Red Knot	Survey occurrence	0.000	0.000	0.000
Scolopacidae	Red Knot	Site occurence	0.000	0.000	0.000
Scolopacidae	Red Knot	Occurrence count	5.443	0.000	5.443
Scolopacidae	Red Knot	Detection rate	0.815	0.000	0.815
Scolopacidae	Red Phalarope	Survey occurrence	0.000	0.000	0.000
Scolopacidae	Red Phalarope	Site occurence	0.000	0.000	0.000
Scolopacidae	Red Phalarope	Occurrence count	9.400	0.000	9.400
Scolopacidae	Red Phalarope	Detection rate	0.911	0.000	0.911
Scolopacidae	Red-necked Phalarope	Survey occurrence	0.000	0.001	0.000
Scolopacidae	Red-necked Phalarope	Site occurence	0.000	0.001	0.000
Scolopacidae	Red-necked Phalarope	Occurrence count	8.597	1.037	7.560
Scolopacidae	Red-necked Phalarope	Detection rate	0.943	0.129	0.814
Scolopacidae	Rock Sandpiper	Survey occurrence	0.000	0.000	0.000
Scolopacidae	Rock Sandpiper	Site occurence	0.000	0.000	0.000
Scolopacidae	Rock Sandpiper	Occurrence count	2.519	0.000	2.519

Table 8: Metrics of detection occurrence per species for ARU and point count surveys across the waterbird dataset. *(continued)*

Family	**Species**	**metric**	**Point count**	**ARU**	**Difference**
Scolopacidae	Rock Sandpiper	Detection rate	0.997	0.000	0.997
Scolopacidae	Ruddy Turnstone	Survey occurrence	0.000	0.000	0.000
Scolopacidae	Ruddy Turnstone	Site occurence	0.000	0.000	0.000
Scolopacidae	Ruddy Turnstone	Occurrence count	6.469	1.000	5.469
Scolopacidae	Ruddy Turnstone	Detection rate	0.884	0.250	0.634
Scolopacidae	Ruff	Survey occurrence	0.000	0.000	0.000
Scolopacidae	Ruff	Site occurence	0.000	0.000	0.000
Scolopacidae	Ruff	Occurrence count	1.040	0.000	1.040
Scolopacidae	Ruff	Detection rate	0.980	0.000	0.980
Scolopacidae	Sanderling	Survey occurrence	0.000	0.000	0.000
Scolopacidae	Sanderling	Site occurence	0.000	0.000	0.000
Scolopacidae	Sanderling	Occurrence count	13.021	0.000	13.021
Scolopacidae	Sanderling	Detection rate	0.817	0.000	0.817
Scolopacidae	Semipalmated Sandpiper	Survey occurrence	0.001	0.000	0.000
Scolopacidae	Semipalmated Sandpiper	Site occurence	0.001	0.001	0.000
Scolopacidae	Semipalmated Sandpiper	Occurrence count	101.438	1.117	100.000
Scolopacidae	Semipalmated Sandpiper	Detection rate	0.829	0.092	0.737
Scolopacidae	Sharp-tailed Sandpiper	Survey occurrence	0.000	0.000	0.000
Scolopacidae	Sharp-tailed Sandpiper	Site occurence	0.000	0.000	0.000
Scolopacidae	Sharp-tailed Sandpiper	Occurrence count	1.000	0.000	1.000
Scolopacidae	Sharp-tailed Sandpiper	Detection rate	0.526	0.000	0.526
Scolopacidae	Short-billed Dowitcher	Survey occurrence	0.000	0.001	0.000
Scolopacidae	Short-billed Dowitcher	Site occurence	0.000	0.002	-0.002
Scolopacidae	Short-billed Dowitcher	Occurrence count	6.762	1.051	5.711
Scolopacidae	Short-billed Dowitcher	Detection rate	0.814	0.140	0.674
Scolopacidae	Solitary Sandpiper	Survey occurrence	0.002	0.006	-0.004
Scolopacidae	Solitary Sandpiper	Site occurence	0.002	0.018	-0.016
Scolopacidae	Solitary Sandpiper	Occurrence count	1.300	1.043	0.257
Scolopacidae	Solitary Sandpiper	Detection rate	0.863	0.381	0.483
Scolopacidae	Spotted Sandpiper	Survey occurrence	0.007	0.005	0.001
Scolopacidae	Spotted Sandpiper	Site occurence	0.008	0.019	-0.011
Scolopacidae	Spotted Sandpiper	Occurrence count	1.341	1.094	0.247
Scolopacidae	Spotted Sandpiper	Detection rate	0.930	0.476	0.454
Scolopacidae	Stilt Sandpiper	Survey occurrence	0.000	0.001	0.000
Scolopacidae	Stilt Sandpiper	Site occurence	0.000	0.001	-0.001
Scolopacidae	Stilt Sandpiper	Occurrence count	5.388	1.079	4.309
Scolopacidae	Stilt Sandpiper	Detection rate	0.905	0.237	0.668
Scolopacidae	Surfbird	Survey occurrence	0.000	0.000	0.000
Scolopacidae	Surfbird	Site occurence	0.000	0.000	0.000
Scolopacidae	Surfbird	Occurrence count	8.880	1.125	7.755
Scolopacidae	Surfbird	Detection rate	0.888	0.310	0.578
Scolopacidae	Upland Sandpiper	Survey occurrence	0.005	0.003	0.002
Scolopacidae	Upland Sandpiper	Site occurence	0.007	0.012	-0.005
Scolopacidae	Upland Sandpiper	Occurrence count	1.309	1.184	0.125
Scolopacidae	Upland Sandpiper	Detection rate	0.991	0.493	0.498

Table 8: Metrics of detection occurrence per species for ARU and point count surveys across the waterbird dataset. *(continued)*

Family	**Species**	**metric**	**Point count**	**ARU**	**Difference**
Scolopacidae	Wandering Tattler	Survey occurrence	0.000	0.000	0.000
Scolopacidae	Wandering Tattler	Site occurence	0.000	0.000	0.000
Scolopacidae	Wandering Tattler	Occurrence count	1.167	1.027	0.140
Scolopacidae	Wandering Tattler	Detection rate	0.920	0.152	0.769
Scolopacidae	Western Sandpiper	Survey occurrence	0.000	0.000	0.000
Scolopacidae	Western Sandpiper	Site occurence	0.000	0.000	0.000
Scolopacidae	Western Sandpiper	Occurrence count	31.381	0.000	31.381
Scolopacidae	Western Sandpiper	Detection rate	0.868	0.000	0.868
Scolopacidae	Whimbrel	Survey occurrence	0.000	0.002	-0.002
Scolopacidae	Whimbrel	Site occurence	0.000	0.003	-0.003
Scolopacidae	Whimbrel	Occurrence count	1.555	1.259	0.295
Scolopacidae	Whimbrel	Detection rate	0.996	0.585	0.411
Scolopacidae	White-rumped Sandpiper	Survey occurrence	0.000	0.000	0.000
Scolopacidae	White-rumped Sandpiper	Site occurence	0.000	0.000	0.000
Scolopacidae	White-rumped Sandpiper	Occurrence count	8.780	1.000	7.780
Scolopacidae	White-rumped Sandpiper	Detection rate	0.808	0.250	0.558
Scolopacidae	Willet	Survey occurrence	0.005	0.006	-0.001
Scolopacidae	Willet	Site occurence	0.006	0.019	-0.013
Scolopacidae	Willet	Occurrence count	1.400	1.242	0.158
Scolopacidae	Willet	Detection rate	0.985	0.409	0.576
Scolopacidae	Wilson's Phalarope	Survey occurrence	0.002	0.001	0.001
Scolopacidae	Wilson's Phalarope	Site occurence	0.002	0.003	0.000
Scolopacidae	Wilson's Phalarope	Occurrence count	4.460	1.216	3.244
Scolopacidae	Wilson's Phalarope	Detection rate	0.981	0.374	0.607
Scolopacidae	Wilson's Snipe	Survey occurrence	0.033	0.066	-0.033
Scolopacidae	Wilson's Snipe	Site occurence	0.041	0.130	-0.089
Scolopacidae	Wilson's Snipe	Occurrence count	1.258	1.303	-0.046
Scolopacidae	Wilson's Snipe	Detection rate	0.975	0.533	0.442